

R&B：统一test与机制研究总报告

2026_09_06_14_05_08 研究批次 | 完成记录实时核对：2026-09-08

有正向性能信号，有可核验的通信机制；
普遍几何优势和临床推广仍未得到确定证据。

855 / 855

test评价作业接受

21 / 21

投影图谱检查点

121

锁定的主要比较

855项是评价及诊断作业，不是855次新增训练。包含777个模型视图、24项宿主诊断和54项配对诊断。全量原始汇总CSV与逐图源数据另附；本PDF覆盖关键结论和全部31幅已验收公开图。

先看结论：哪些成立，哪些还不能确定

问题	综合判断
R&B是否有用？	参考配置test分支均值60.73% / 61.73%，无通信58.13%；两分支均有正向均值变化，但不能据此宣称普遍显著。
是否胜过普通通信？	SVD/QR匹配组几何收益约+1.46 / +1.43 pp；全局同时区间均跨0，收益大小仍不确定。
空间结构是否必要？	空间打乱有时更高；S轴邻接对照也不确定。尚未证明完整Radon几何是收益不可替代的来源。
SVD或更大宽度一定更好？	未成立。test中中心化SVD也有较高均值；QR三档均值递增，但不构成一般单调性保证。
多桥、冻结、宿主增强呢？	多桥没有明确额外收益；冻结组仍有正向信号；宿主增强训练约+2 pp，但区间仍宽。
最明确的功能证据？	选中宿主+R&B模型关闭R&B后显著受损，说明功能依赖。约14 pp删除效应不能写成新增训练收益。
现在应做什么？	承接固定机制假设扩大数据和骨干；审计标签与数据角色。继续扫描这批test不能建立独立证据。

Completed evidence and scope

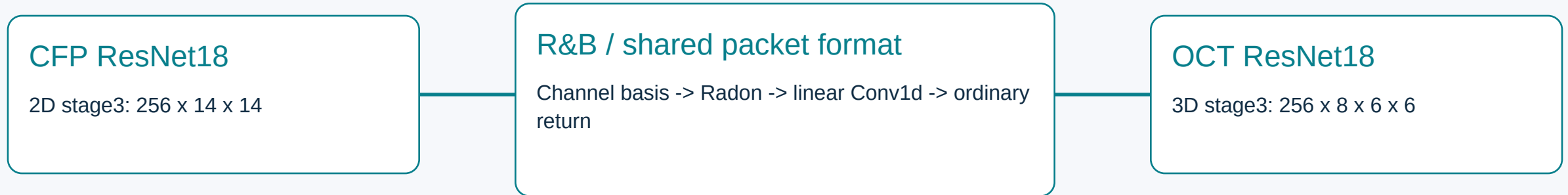
Reference rows, unique model views and inference jobs are different counting units.

Evidence source	Reference rows / jobs	Role
Current direct + host + original frozen	384 rows	342 direct + 36 augmentation + 6 frozen
Historical final213	231 rows	9 joint-width trainings expand to 27 evaluation views
Historical S-axis adjacency	18 rows	Ordered versus permuted S-axis
Mechanism completion	72 rows	24 fixed-r geometry + 48 frozen completion
Multi-depth bridge study	96 rows	Single / double / triple bridge controls
Independent parent checkpoints	3 rows	Original unimodal models
After model-view deduplication	804 rows -> 777 views	Same checkpoint + inference configuration reused
Unified test queue	777 + 24 + 54 = 855 jobs	All accepted; zero failed / active / pending
Post-test descriptive diagnostics	21 atlas; 22 synthetic geometries	21 synthetic feasible; 1 dense-operator guard infeasible

The synthetic infeasible geometry is a technical limit, not a poor performance result. Historical evidence is retained.

Two task networks, one structured residual bridge

Fixed-subspace formulation shown. Stage3 reference: M32 / S64 / k3; native networks remain nonlinear.



$$\begin{aligned}
 z_i &= A_i Q_i^T X_i \\
 u_i &= \sum_j K_{ij} * z_j \\
 Y_i &= X_i + Q_i B_i u_i
 \end{aligned}$$

Two heads retain CE_CFP + CE_OCT and two predictions. Main metric = mean of their macro-F1 values.

Fixed Q, geometry and permutations are buffers. Only the central bridge convolution is learned in fixed-basis arms.

Full adaptation updates both networks and BN. Frozen arms hold backbone, head and BN fixed; only the bridge is trained.

How strong is the statistical evidence?

Use the locked comparison weights. Report ordinary and simultaneous intervals, not selected test winners.

Global decision over 121 primary comparisons	Count	Meaning
Supports substantive improvement	3	All three are within-checkpoint functional switches
Supports practical similarity	3	Entire interval lies inside the research band +/-1 pp
Unresolved	114	Precision does not distinguish the prespecified categories
Not estimable	1	No estimable matched terms; retained explicitly

10,000 participant-level paired bootstrap resamples; 290 test participants. Models, seeds, widths and permutation repeats do not increase this sample size.

Global max-|t| approximate simultaneous intervals cover all primary comparisons; family and ordinary intervals are also retained.

The +/-1 pp band is a research convention, not a clinical threshold. Failure to reject is not proof of equivalence.

Development selected checkpoints and configurations. Earlier LOOK use of this cohort is recorded: this is not a pristine, never-touched historical test or external clinical validation.

Absolute performance: development is not test

Three seeds x two backbone learning rates; equal weight per model. All values are percentages.

Configuration	Dev mean F1	Test CFP	Test OCT	Test mean F1	Fixed prob. fuse
No communication	65.03	59.09	57.17	58.13	59.66
Ordinary r16 / h512	65.10	59.50	57.25	58.37	59.78
R&B r16 / h512	67.67	60.61	60.86	60.73	62.33
Ordinary r32 / h1024	65.57	59.23	58.18	58.71	60.59
R&B r32 / h1024	68.24	62.81	60.66	61.73	61.82
MMTM hidden128	65.02	58.91	57.54	58.23	59.49
MMTM hidden256	65.02	58.91	57.54	58.23	59.49
MMTM hidden512	65.02	58.91	57.54	58.23	59.49
MMTM hidden1024	65.02	58.91	57.54	58.23	59.49
Attention d128	65.41	59.63	57.20	58.42	60.08
Attention d256	65.47	59.50	58.58	59.04	60.08
Attention d512	66.16	59.81	59.10	59.45	60.78
Attention d1024	66.18	61.20	60.44	60.82	60.58

Reference R&B improves both branch means. Absolute test performance is lower; it should not be reported as the earlier ~70% development result.

Cost and baseline interpretation

Identical training rules do not imply equal parameter counts, computation or historical search budgets.

Configuration	Total params M	Added params M	Peak reserved GiB	Mean train min
No communication	44.34	0.00	7.19	3.05
R&B r16 / h512	47.48	3.15	7.30	6.58
R&B r32 / h1024	56.92	12.58	7.40	5.01
MMTM hidden256	44.60	0.26	7.32	2.22
Attention d256	44.87	0.53	7.55	3.86
Attention d1024	46.44	2.10	7.76	2.95

All 24 current MMTM runs (4 widths x 6 models) selected epoch 0. Their equal test scores reflect rollback to the initial model, not proof that channel width is intrinsically irrelevant.

R&B is linear inside the fixed bridge, but the h512/h1024 references add more parameters than these MMTM and attention configurations. Do not claim that linear automatically means cheaper.

Training wall time includes differing plateau lengths and shared-machine load. This table is accounting, not a controlled latency benchmark.

Peak reserved memory is a per-process training measurement; project caps and concurrent processes were monitored separately.

Compression and width: no universal winner

Three seeds, backbone LR6e-5, M32 / S64 / k3. Test branch-mean F1: mean (sample SD over three seeds), in percentage points.

Method	rho1/16: r16 h512	rho1/8: r32 h1024	rho1/4: r64 h2048
Learned CM projection	60.12 (1.39)	59.28 (1.84)	60.20 (2.19)
Learned channel projection	60.26 (1.63)	61.57 (0.87)	60.44 (0.82)
Uncentered SVD + Radon	61.14 (4.07)	61.53 (0.95)	59.53 (0.35)
Centered SVD + Radon	61.25 (1.85)	62.08 (1.66)	60.27 (0.86)
Random QR + Radon	59.15 (1.65)	60.18 (1.01)	60.97 (0.46)
SVD + ordinary	58.52 (0.87)	58.76 (0.59)	60.52 (0.65)
SVD + self-only	58.23 (0.82)	58.23 (0.82)	58.23 (0.82)
SVD + scrambled	60.76 (0.97)	62.28 (2.58)	60.96 (1.69)
QR + scrambled	59.23 (0.33)	61.44 (0.31)	61.99 (1.58)

Centered SVD also has higher observed means; scrambled controls can exceed standard Radon. QR rises across these three means, but this is not a monotonicity guarantee.

What remains uncertain about Radon geometry

Differences are percentage points; intervals below are global simultaneous 95% intervals.

Locked comparison	Test difference	Global 95% interval
SVD: Radon - ordinary	1.46	[-2.18, 5.11]
QR: Radon - ordinary	1.43	[-1.43, 4.29]
SVD geometry gain - QR geometry gain	0.03	[-2.70, 2.76]
QR: Radon - self-only	1.87	[-1.49, 5.24]
QR: Radon - spatially scrambled	-0.79	[-2.96, 1.39]
SVD: ordered S - permuted S	0.21	[-2.17, 2.59]
QR: ordered S - permuted S	-0.53	[-2.53, 1.47]
Equal-compute geometry gain: budget512	2.92	[-0.26, 6.11]
Equal-compute geometry gain: budget1024	2.83	[-0.47, 6.12]

Positive geometry means are encouraging. Current evidence does not establish that intact Radon geometry or S-axis adjacency is uniquely necessary.

Directions, rank and bridge count

Fixed r varies h with M; fixed h trades channel rank against directions. These answer different questions.

Locked comparison	Test difference pp	Global 95% interval
r16: geometry gain M32 - M16	1.72	[-0.97, 4.41]
r16: geometry gain M64 - M16	1.95	[-1.44, 5.34]
r32: geometry gain M32 - M16	0.03	[-3.14, 3.21]
r32: geometry gain M64 - M16	-0.41	[-3.46, 2.64]
Radon rank32 - rank16, averaged across M	1.06	[-0.98, 3.10]
Geometry gain: 2 bridges - 1 bridge	-0.08	[-2.23, 2.07]
Geometry gain: 3 bridges - 2 bridges	0.09	[-3.17, 3.34]
Radon 2 bridges - single wider r23	-1.56	[-4.81, 1.69]
Radon 3 bridges - single wider r28	-1.86	[-6.46, 2.74]

No established rule that more angles, more channels or more bridges always helps. Wider single bridges remain an essential capacity control.

Does the bridge require backbone co-adaptation?

Frozen backbone, head and BN; bridge convolution alone is trained. Three seeds x three rho levels.

Frozen comparison	Test difference pp	Global 95% interval
SVD: Radon - ordinary	1.56	[-1.37, 4.48]
SVD: Radon - self-only	1.79	[-1.00, 4.58]
SVD: Radon - original parent	2.11	[-1.01, 5.23]
QR: Radon - ordinary	0.86	[-0.93, 2.64]
QR: Radon - self-only	0.91	[-0.79, 2.61]
QR: Radon - original parent	0.91	[-0.85, 2.68]
SVD geometry gain: full - frozen	-0.09	[-4.49, 4.31]
QR geometry gain: full - frozen	0.58	[-2.87, 4.03]

Frozen models retain positive average signals. The intervals do not establish either that co-adaptation is necessary or that frozen training is equally effective.

Training benefit is not the deletion effect

All rows use selected checkpoints, but the first four compare training procedures and the last three toggle one model.

Comparison	Difference pp	Global 95% interval
Train MMTM + R&B versus host continuation	1.97	[-1.37, 5.31]
Train attention + R&B versus host continuation	1.95	[-0.88, 4.79]
MMTM: add R&B versus add ordinary	2.01	[-1.17, 5.20]
Attention: add R&B versus add ordinary	1.30	[-1.07, 3.67]
Selected MMTM model: both on - host only	14.13	[8.88, 19.37]
Selected attention model: both on - host only	14.14	[8.96, 19.32]
Selected attention model: both on - R&B only	3.60	[1.14, 6.06]

The ~14 pp effect establishes functional dependence after joint adaptation. Actual augmentation training benefits are ~2 pp with unresolved global intervals.

Correct pairing and interpretable message flow

54 standard Radon checkpoints: 2 bases x 3 directions x 3 rho x 3 seeds. Permutations move both eyes together.

Original pairing minus perturbation	Mean drop pp	Range across 54 models
Disable OCT -> CFP	2.75	[-1.50, 14.50]
Disable CFP -> OCT	5.05	[0.00, 14.88]
Disable all cross-source blocks	7.81	[0.00, 27.18]
Shuffle OCT -> CFP	0.72	[-0.73, 3.23]
Shuffle CFP -> OCT	1.25	[-0.25, 4.64]
Shuffle both directions	1.98	[-0.98, 5.87]

These are secondary descriptive summaries: average 20 shuffles within each checkpoint, then average checkpoints. No new significance test is introduced.

Original pairing performs better on average, but individual checkpoints can reverse the pattern. Deletion is a stronger intervention than mismatching a sender.

Sinograms, source-block spectra and returned residuals show where the computation changes; they do not alone identify lesions or prove causality.

What the Fourier and sinogram analyses add

21 atlas checkpoints; initial construction and selected state; complete split aggregates plus fixed limited probes.

Question	Observation / evidence	Boundary
Is projection mathematically interpretable?	Synthetic trajectories; continuous multilinear Fourier reference; moment errors.	Finite quadrature is approximate. 21/22 synthetic configurations feasible.
Which frequencies are emphasized?	120 source kernel blocks; some k3 kernels emphasize low frequencies.	Ordinary communication can look similar; kernel gain is not task gain.
Does a message remain a sinogram?	Input projection and mixed messages have different structure.	Arbitrary source/direction mixing need not satisfy Radon consistency.
What does return do?	Ordinary BA has a spatially spread point response.	B is not a precise inverse; sharp reconstruction is not the training objective.
Do wider bridges retain more signal?	Energy retention and residual size can rise with width.	Retained energy and larger residuals do not imply improved classification.

What to carry into the next dataset and backbone

The current cohort supports a controlled mechanism benchmark, not a complete clinical validation claim.

Retain the core method and controls: R&B, matched ordinary communication, self-only, spatial/S-axis controls, and a strong nonlinear baseline under the same adaptation scenario.

Audit the expanded cohort before training: label provenance, acquisition timing, pairing, exclusions, age/sex/center and split overlap. Current test cannot tune the next protocol.

Prespecify a small set of primary questions and separate confirmation from exploration. More configurations on 290 test people do not create more independent evidence.

Evaluate retrained cross-dataset applicability separately from external validation of a fixed model. Repeat the numerical and resource checks after changing feature shapes.

Keep branch-mean F1 and fixed probability-fusion F1 separate. Calibrated risk and predictive values from this balanced case-control cohort are not population estimates.

Technical atlas and complete comparison appendix

All 31 accepted figures follow. Line/interval PDFs retain vectors; heatmaps use original high-resolution PNGs to avoid PDF seams.

Pages	Content	How to read
17 - 24	All primary test comparison plots (8)	Thick ordinary intervals; thin global simultaneous intervals.
25 - 38	Sinogram atlas (14)	Synthetic geometry, full-cohort message/return maps, widths, bases and depths.
39 - 47	Fourier and BA diagnostics (9)	Point returns, numerical slice discrepancy and matrix-valued kernel gain.
Complete numeric sources: all_references_development_test.csv (804 references), all_model_metrics.csv (777 unique model views per split), and the full contrast JSON/CSV files.		

This synthesis adds only transparent aggregate means and descriptive tables. Original models, accepted predictions, primary comparison weights and all bootstrap intervals are unchanged.

Restricted participant images, predictions, features and permutation arrays remain on ws02. Public figures contain synthetic fields or cohort-level aggregates.

Figure 01 | test primary 01

unified_test / test_primary_01

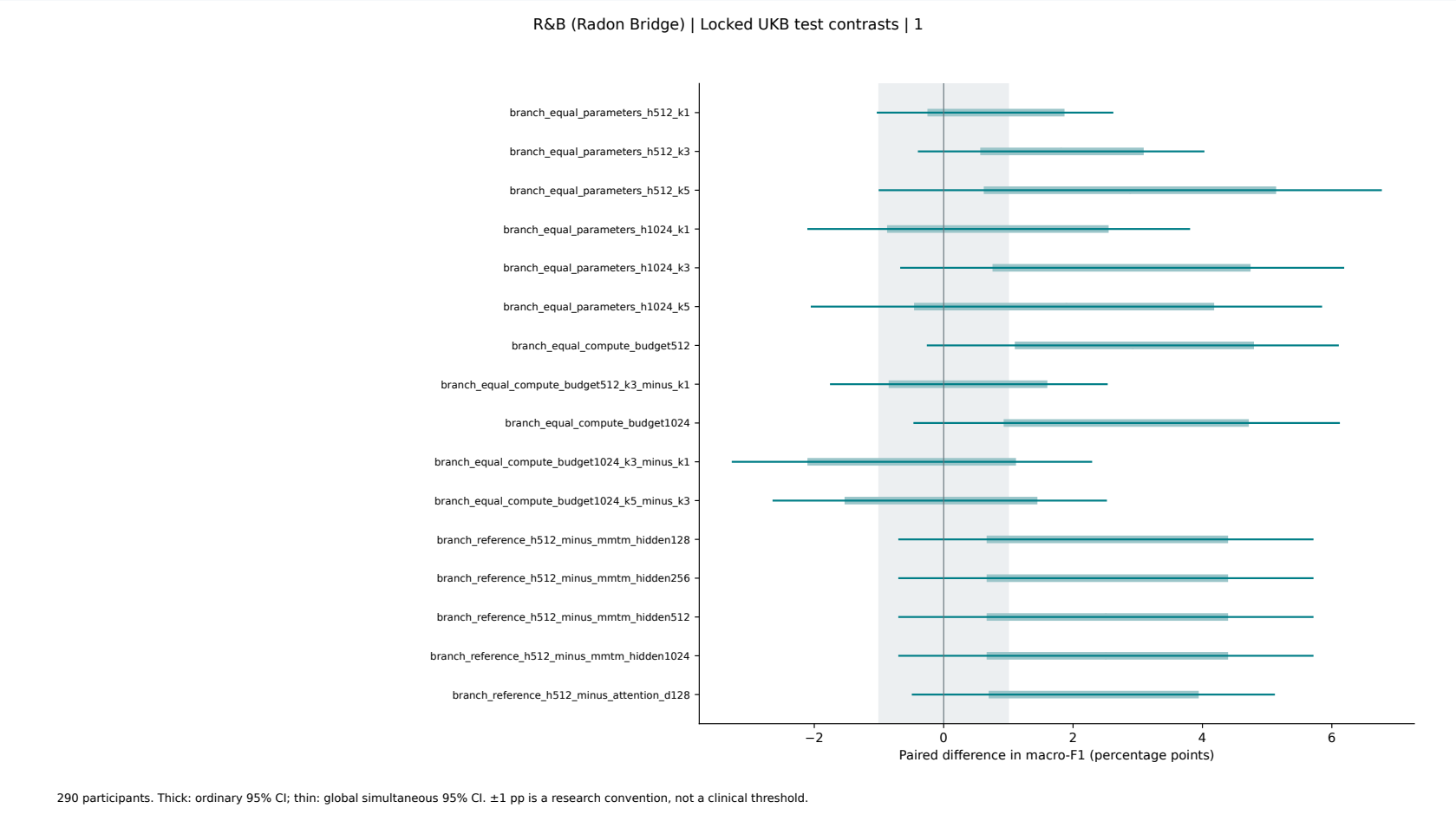


Figure 02 | test primary 02

unified_test / test_primary_02

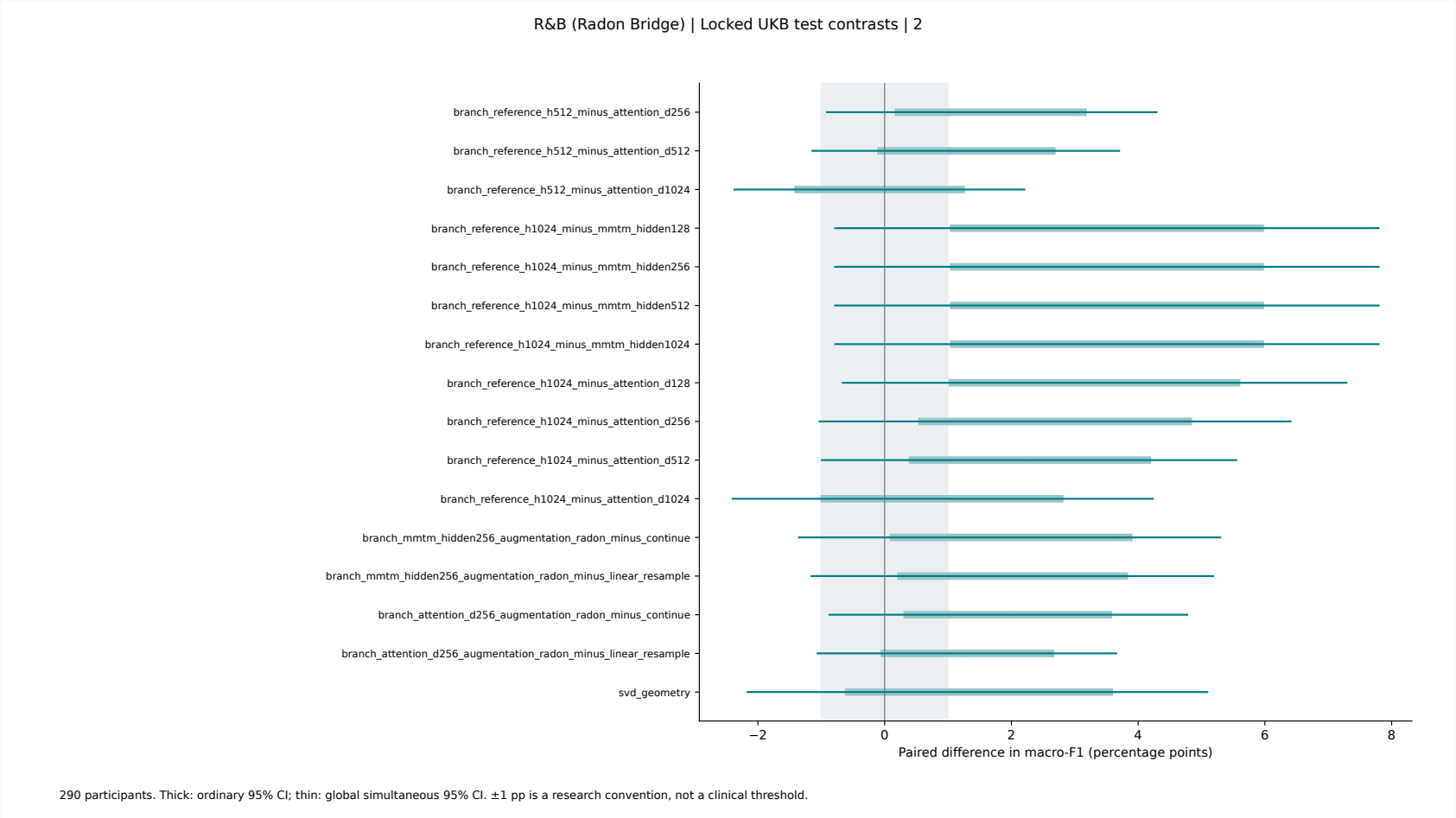


Figure 03 | test primary 03

unified_test / test_primary_03

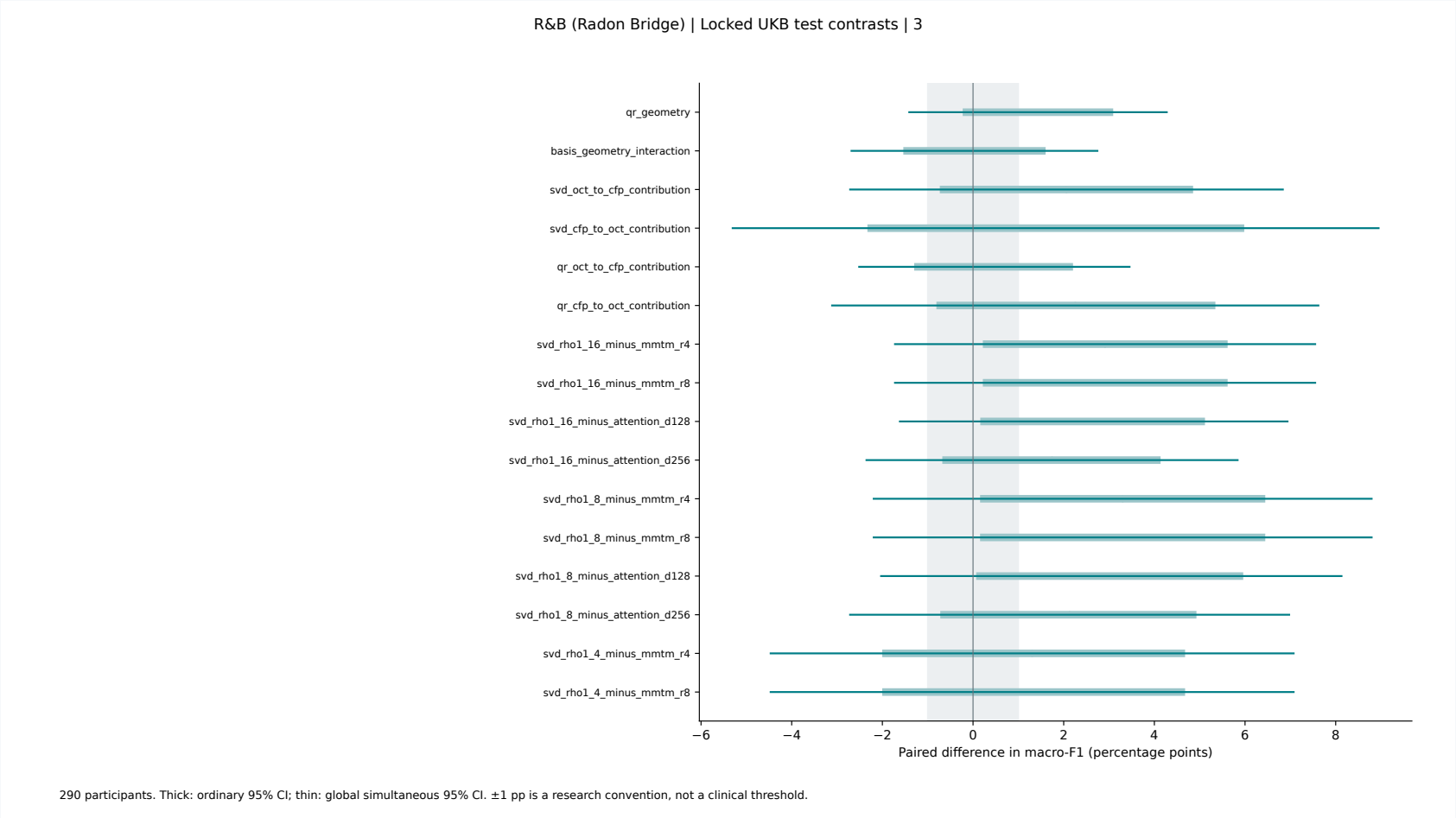


Figure 04 | test primary 04

unified_test / test_primary_04

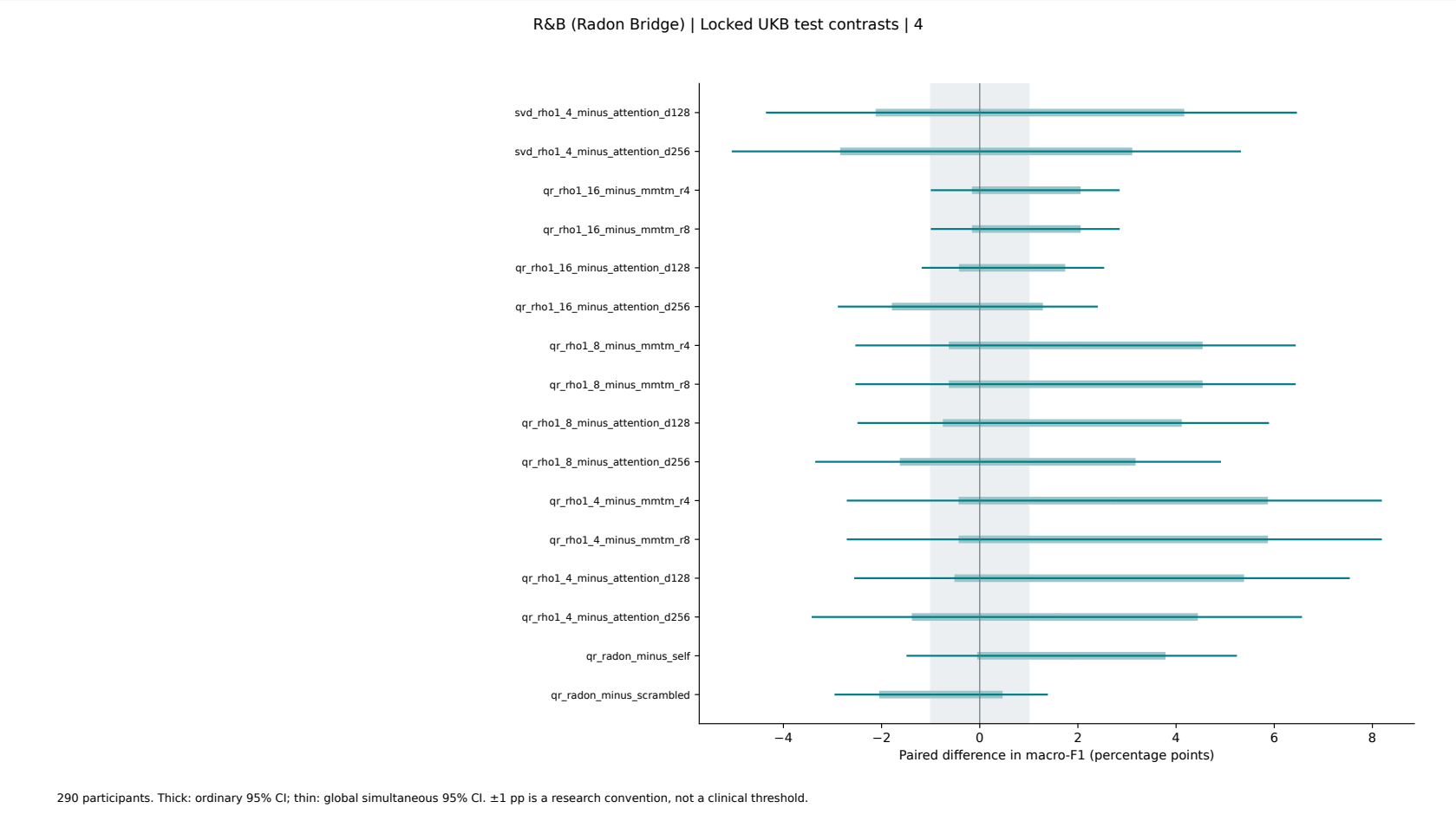


Figure 05 | test primary 05

unified_test / test_primary_05

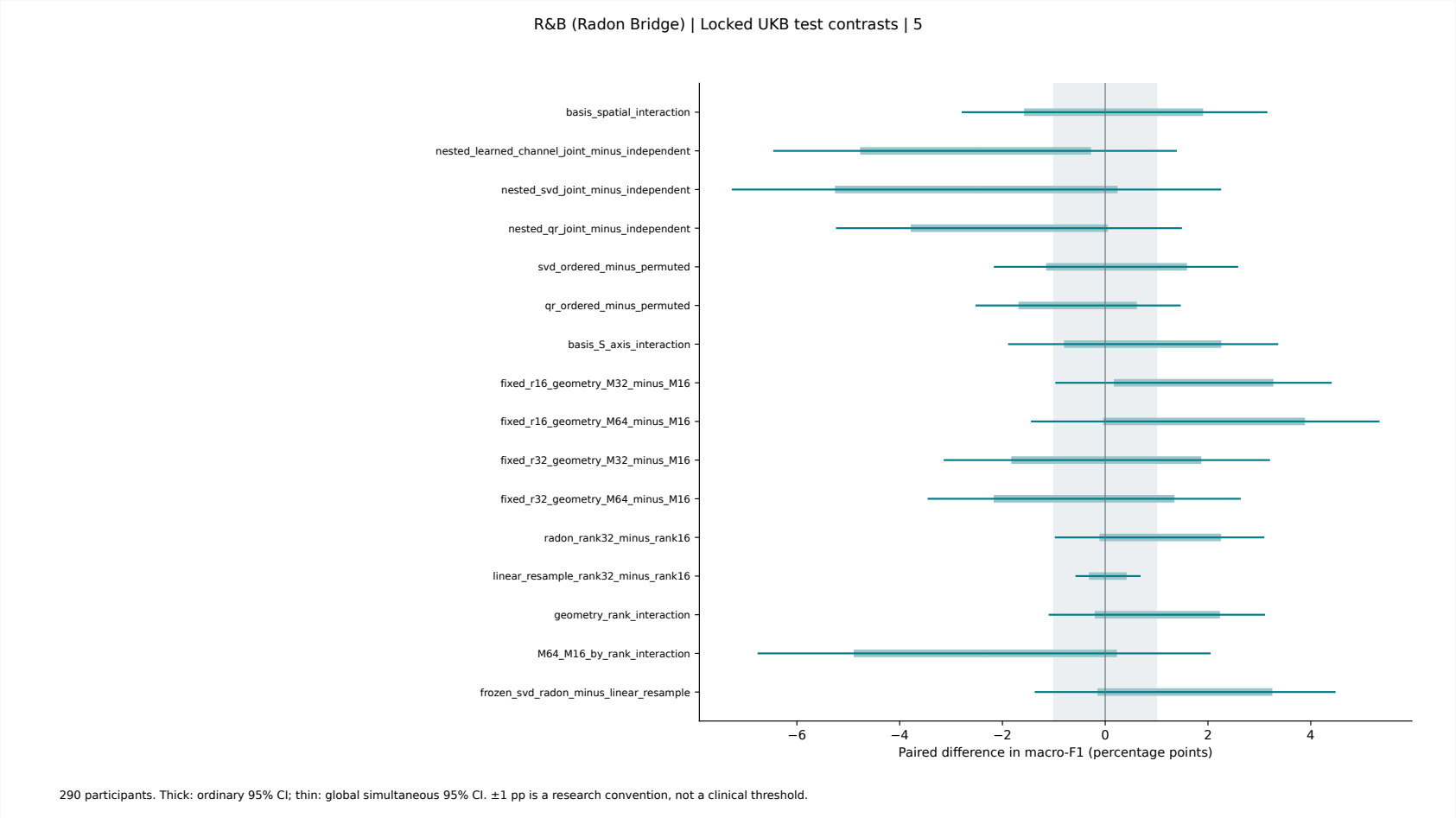


Figure 06 | test primary 06

unified_test / test_primary_06

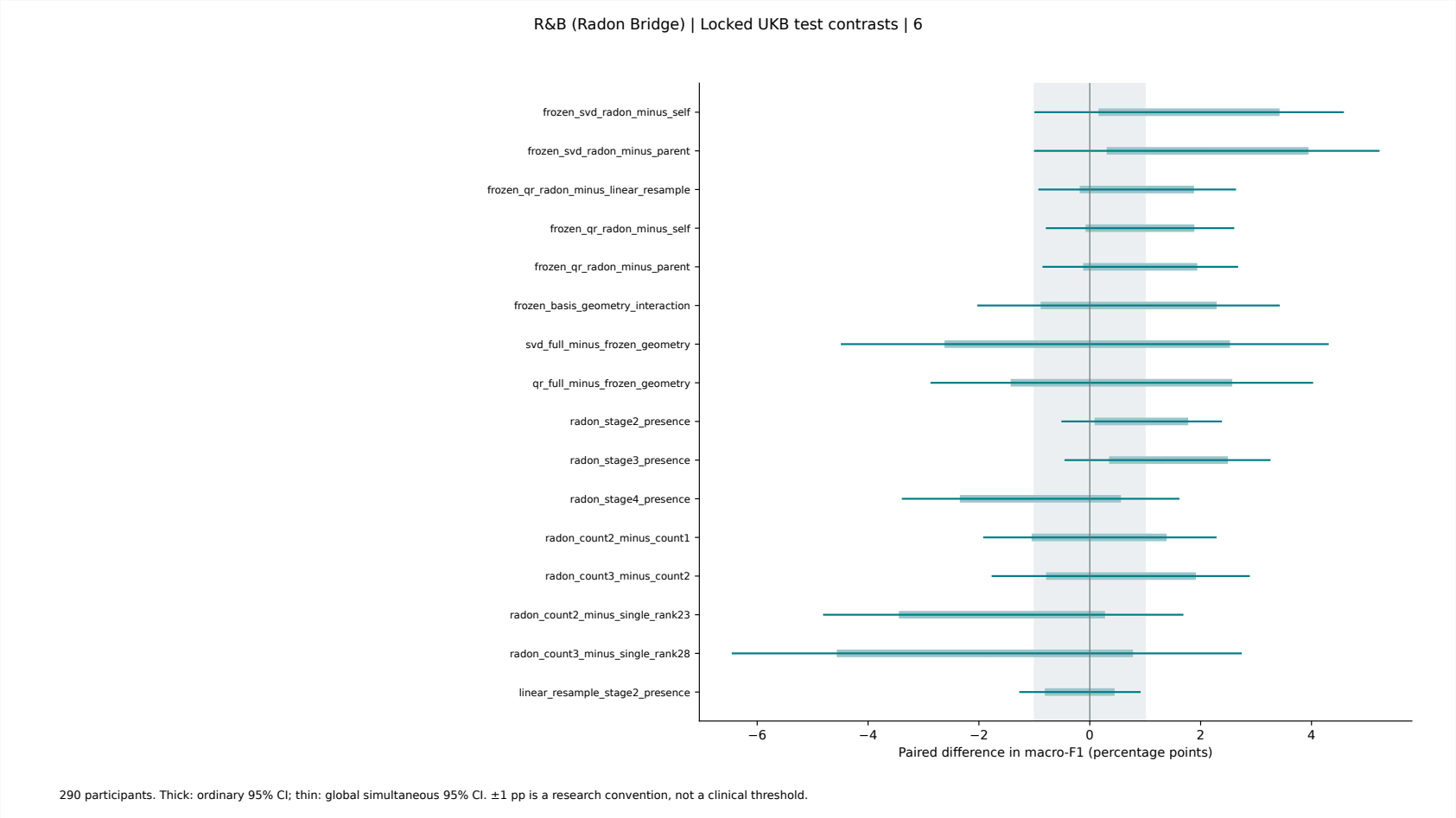


Figure 07 | test primary 07

unified_test / test_primary_07

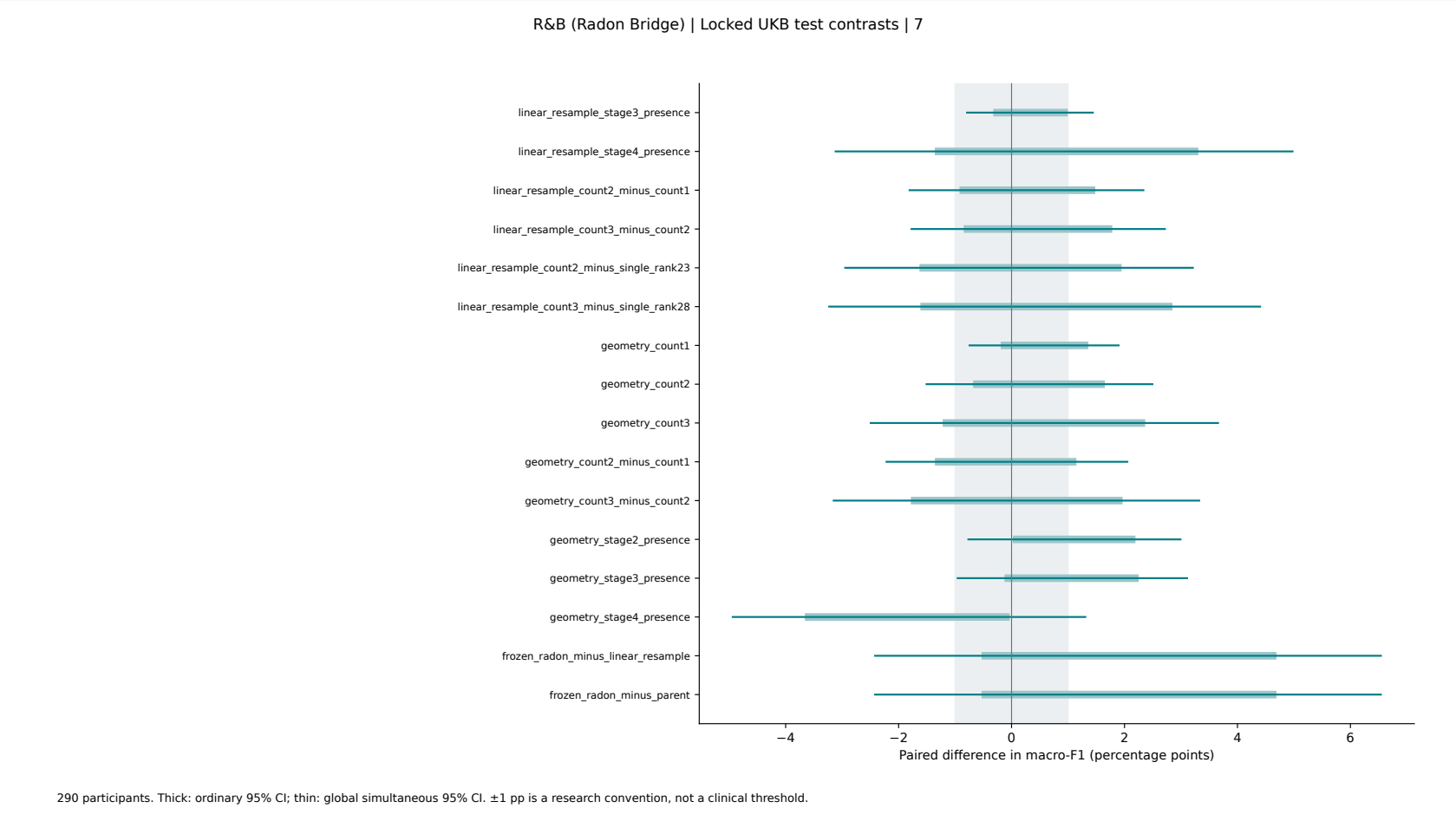


Figure 08 | test primary 08

unified_test / test_primary_08

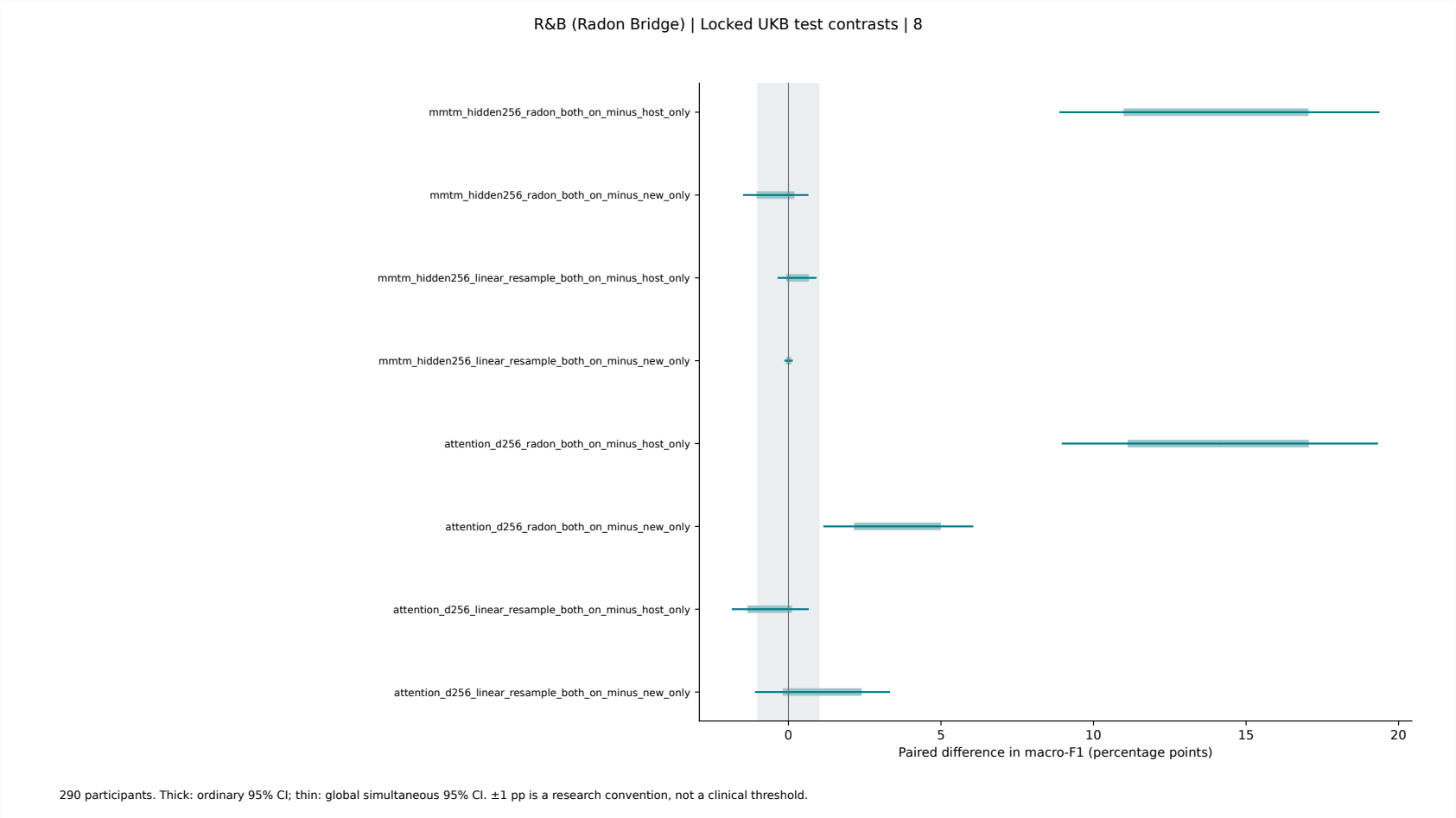


Figure 09 | 01 synthetic 2d

sinogram_atlas / 01_synthetic_2d

R&B | 2D projection geometry on synthetic data

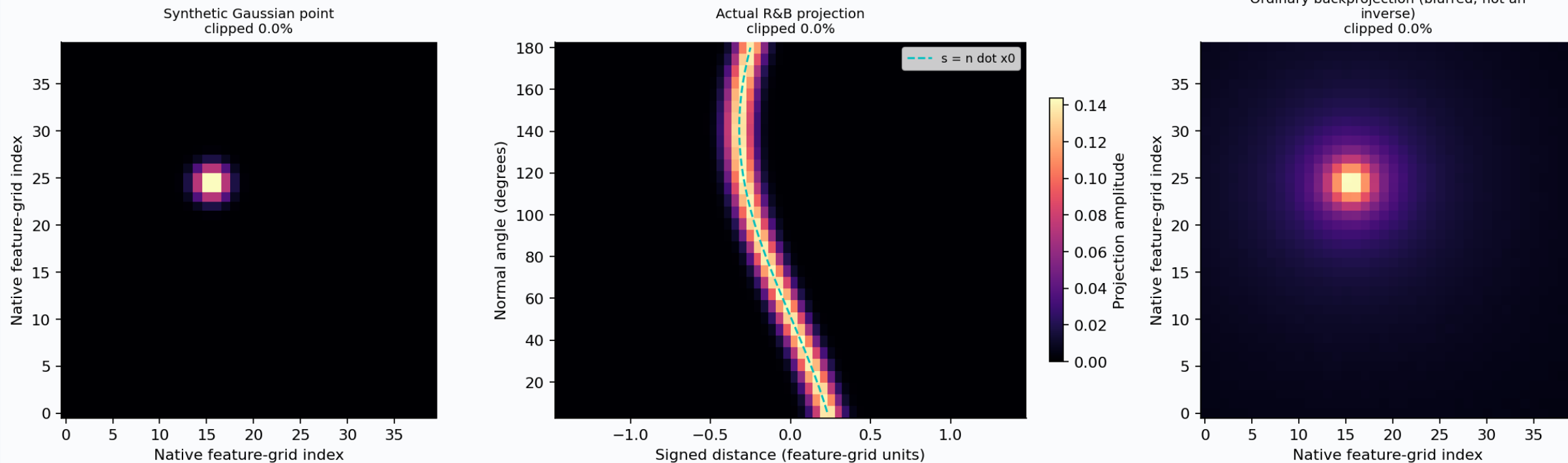


Figure 10 | 01 synthetic 3d

sinogram_atlas / 01_synthetic_3d

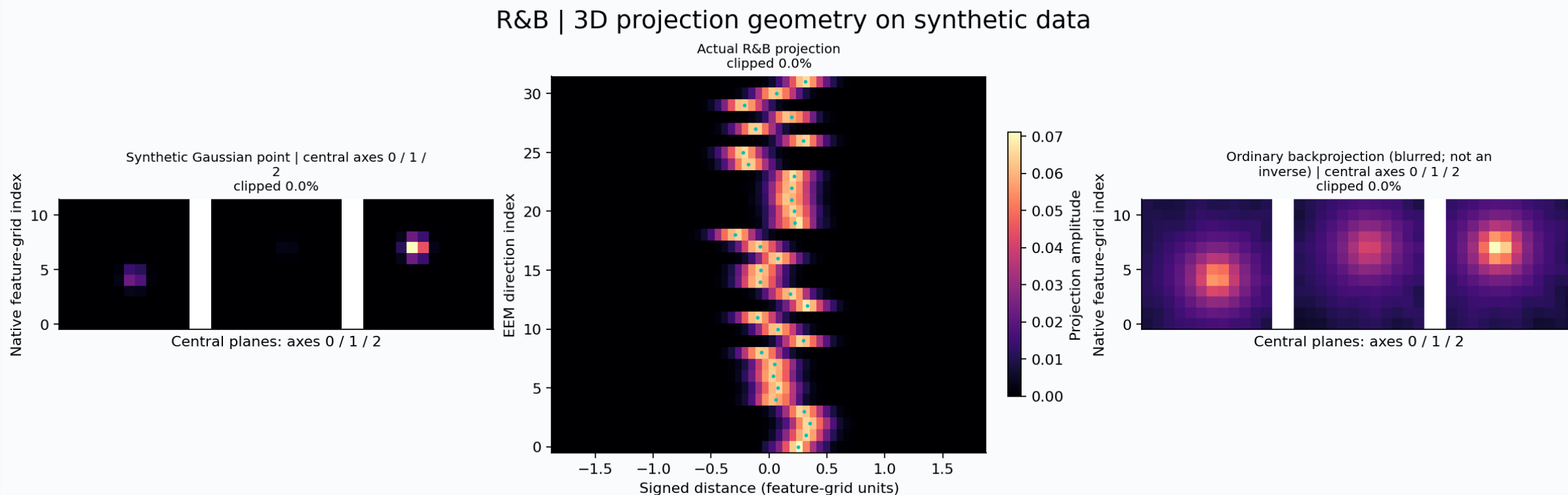


Figure 11 | 02 projected flow cfp stage3

sinogram_atlas / 02_projected_flow_cfp_stage3

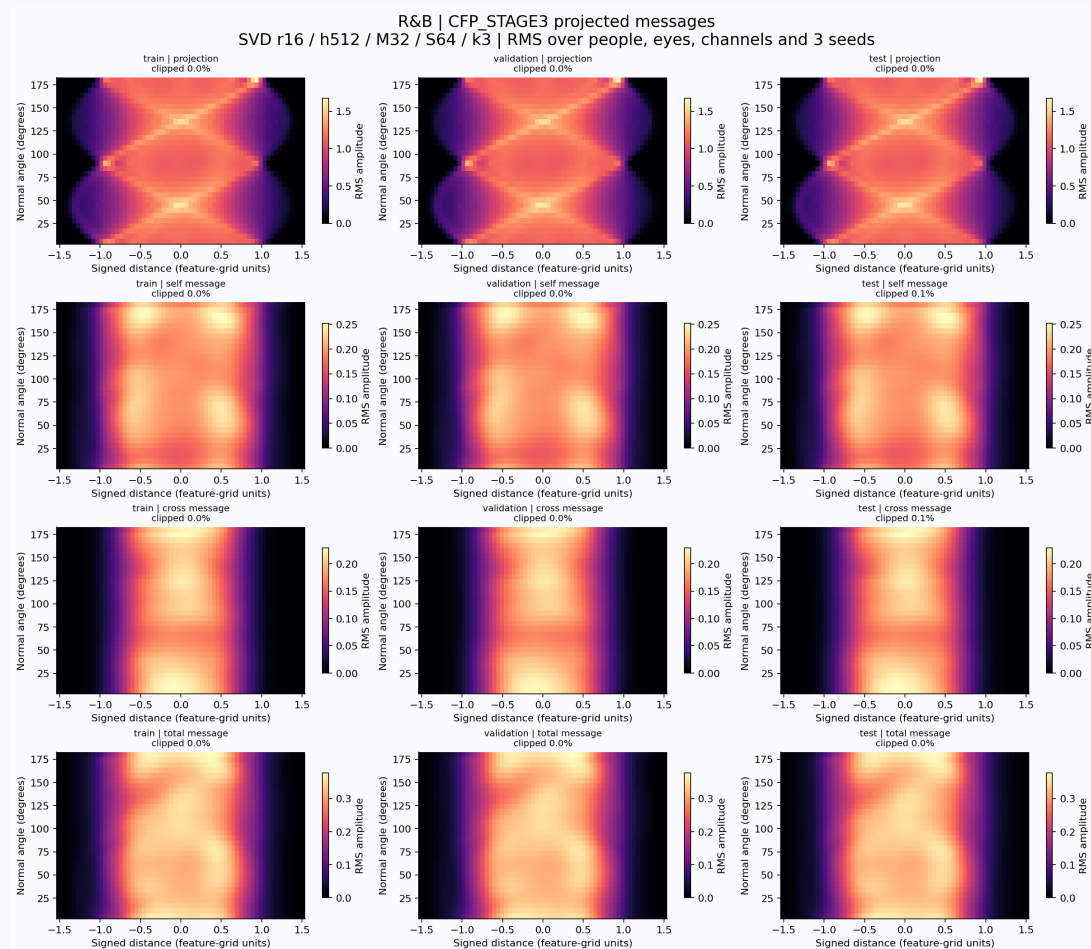


Figure 12 | 02 projected flow oct stage3

sinogram_atlas / 02_projected_flow_oct_stage3

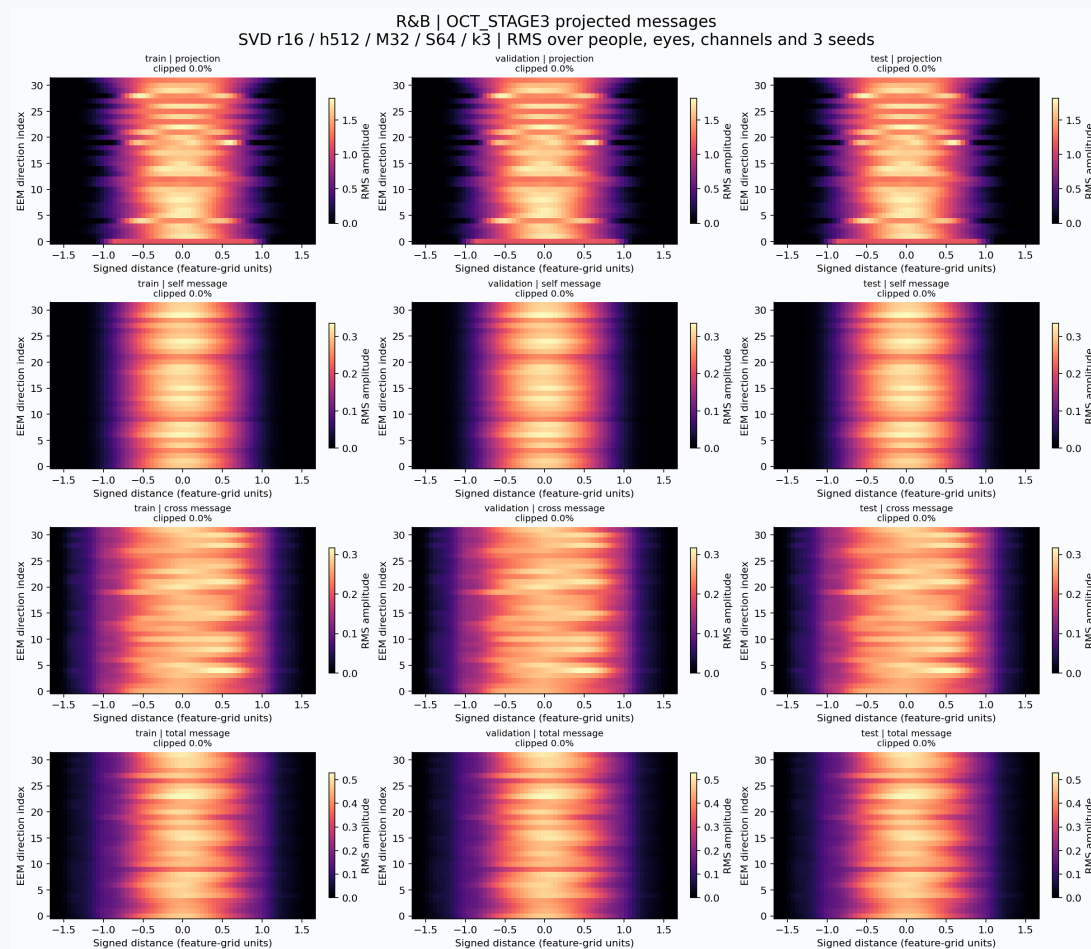


Figure 13 | 03 return maps cfp stage3

sinogram_atlas / 03_return_maps_cfp_stage3

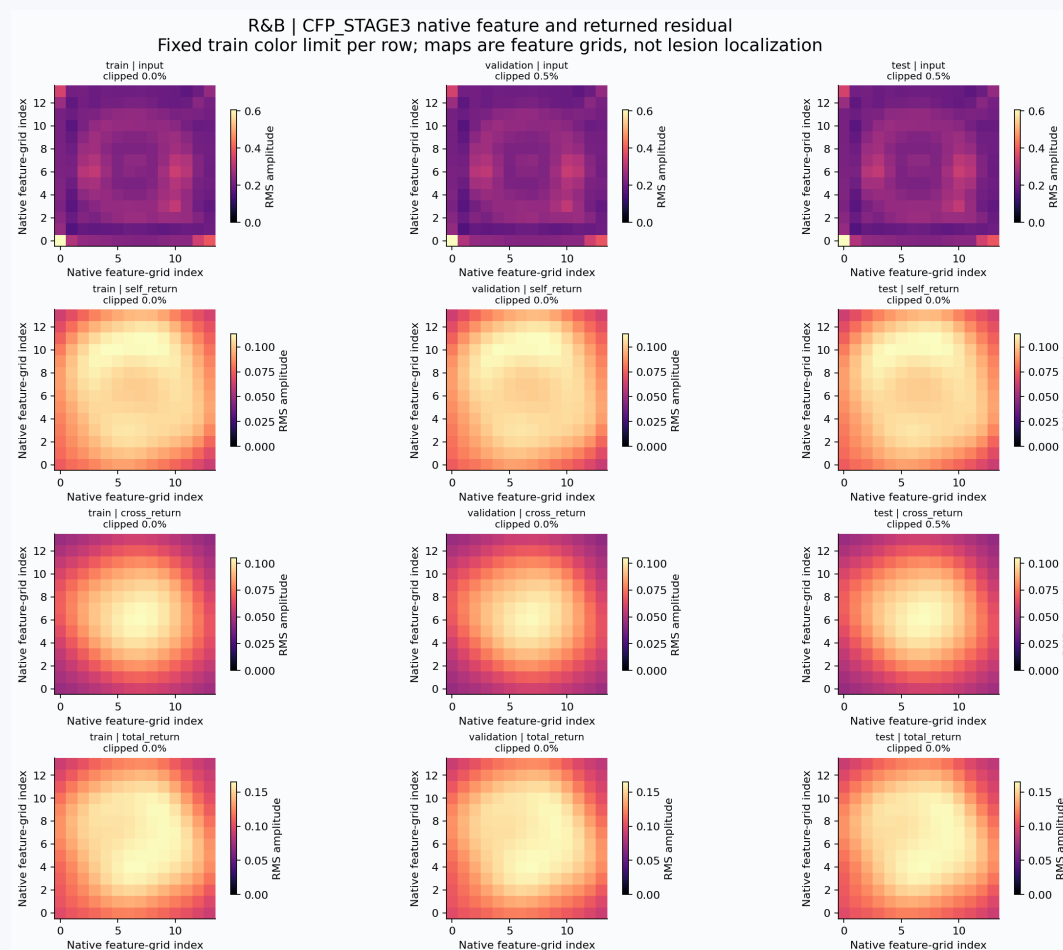


Figure 14 | 03 return maps oct stage3

sinogram_atlas / 03_return_maps_oct_stage3

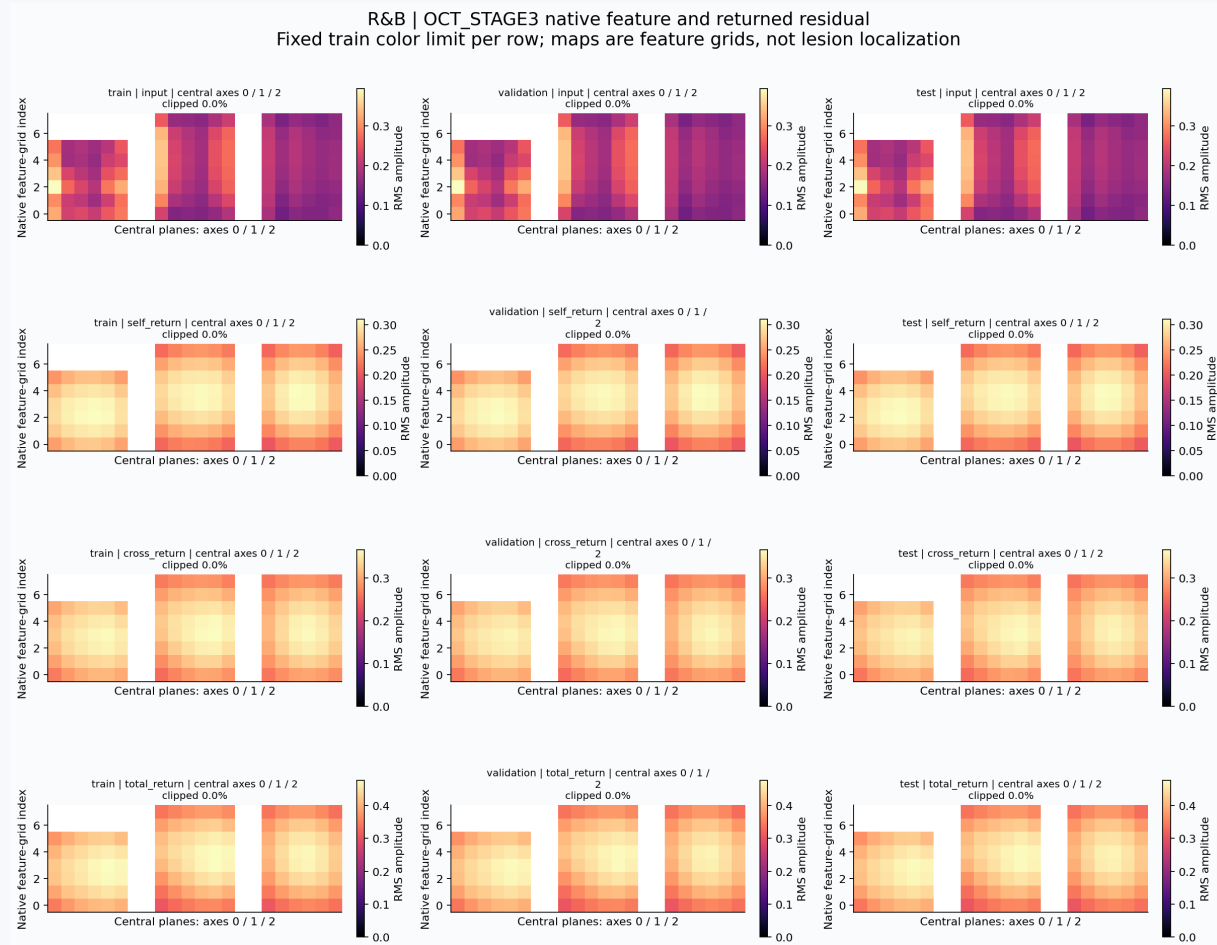


Figure 15 | 04 label difference cfp stage3

sinogram_atlas / 04_label_difference_cfp_stage3

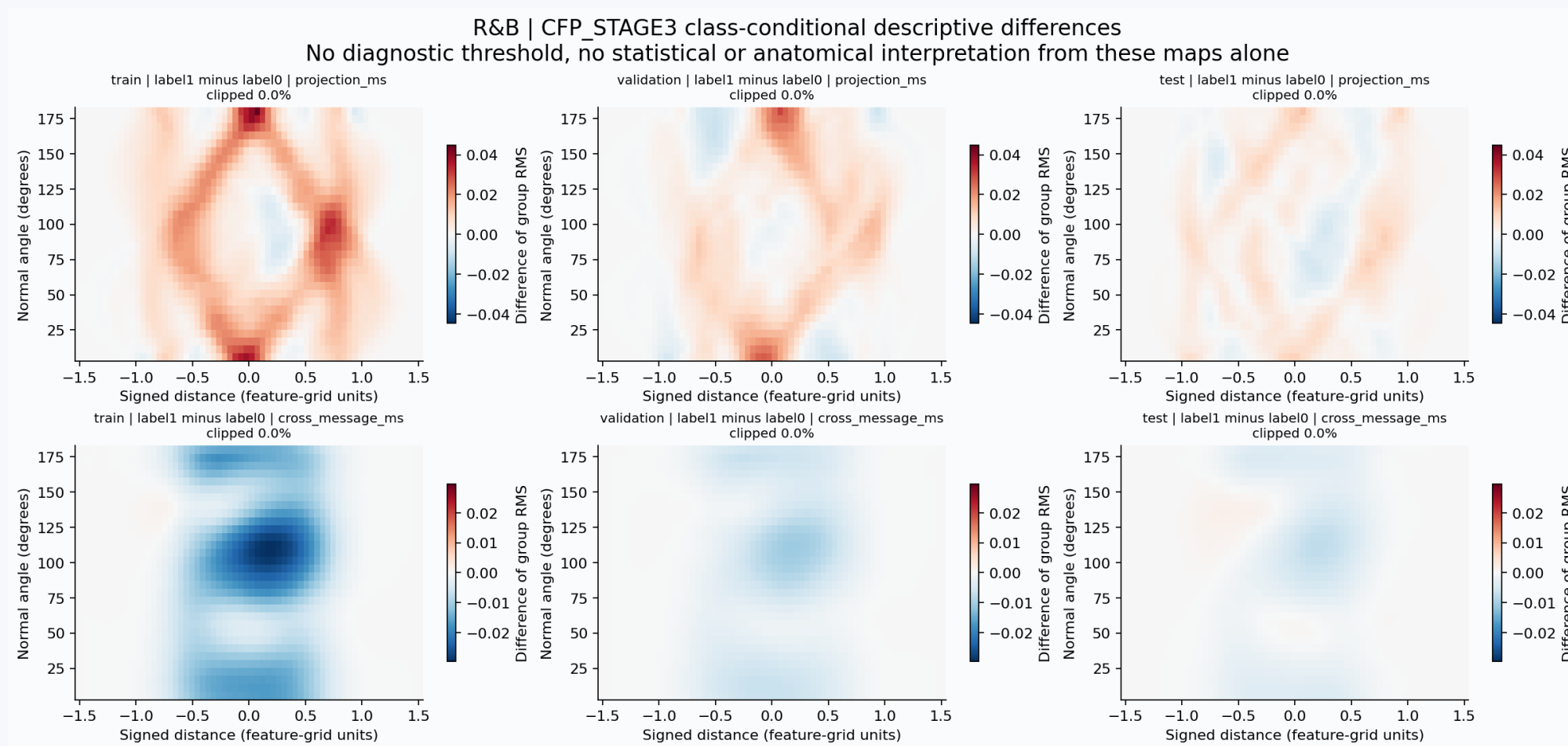


Figure 16 | 04 label difference oct stage3

sinogram_atlas / 04_label_difference_oct_stage3

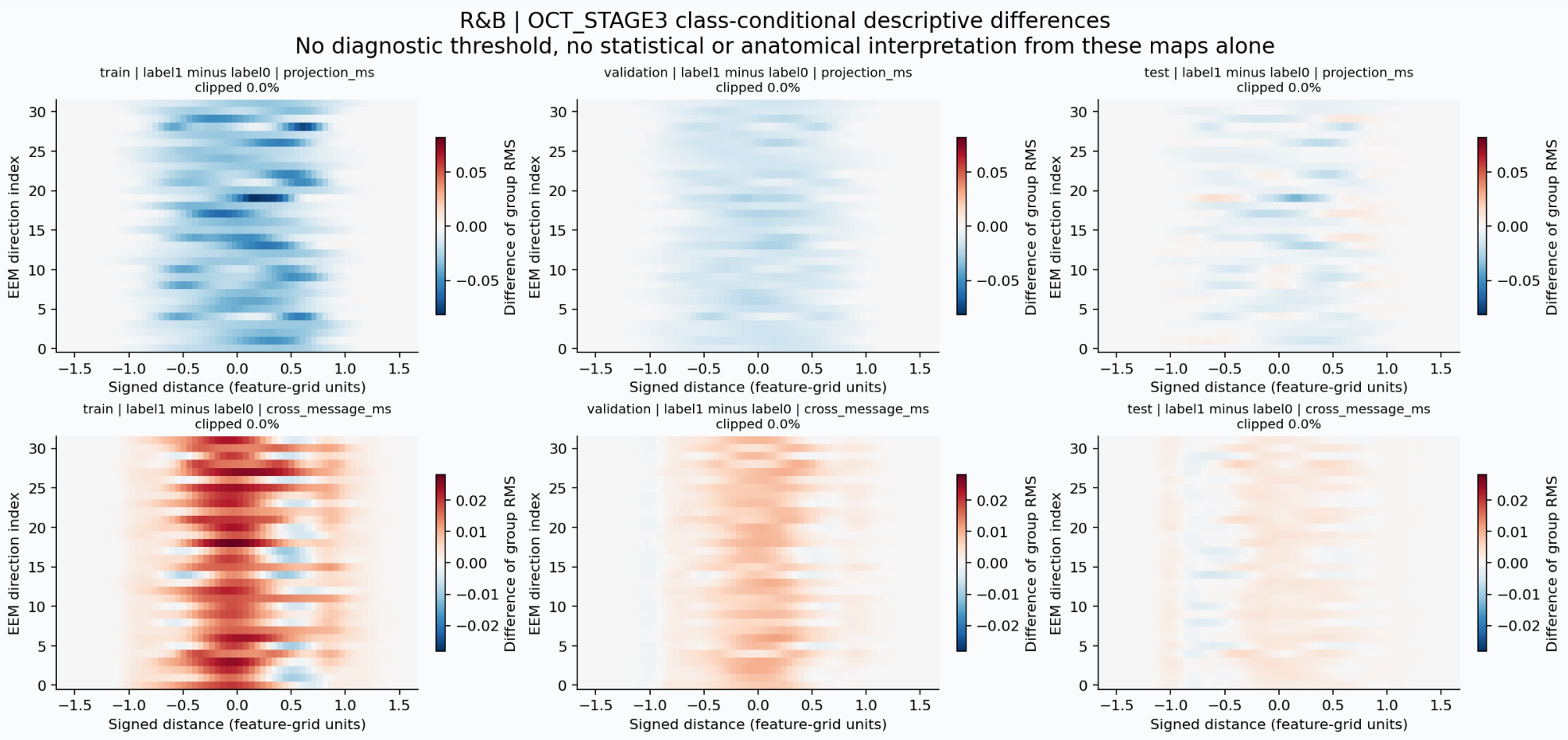


Figure 17 | 05 oct spherical directions

sinogram_atlas / 05_oct_spherical_directions

R&B | OCT directions occupy a sphere, not a single angle axis

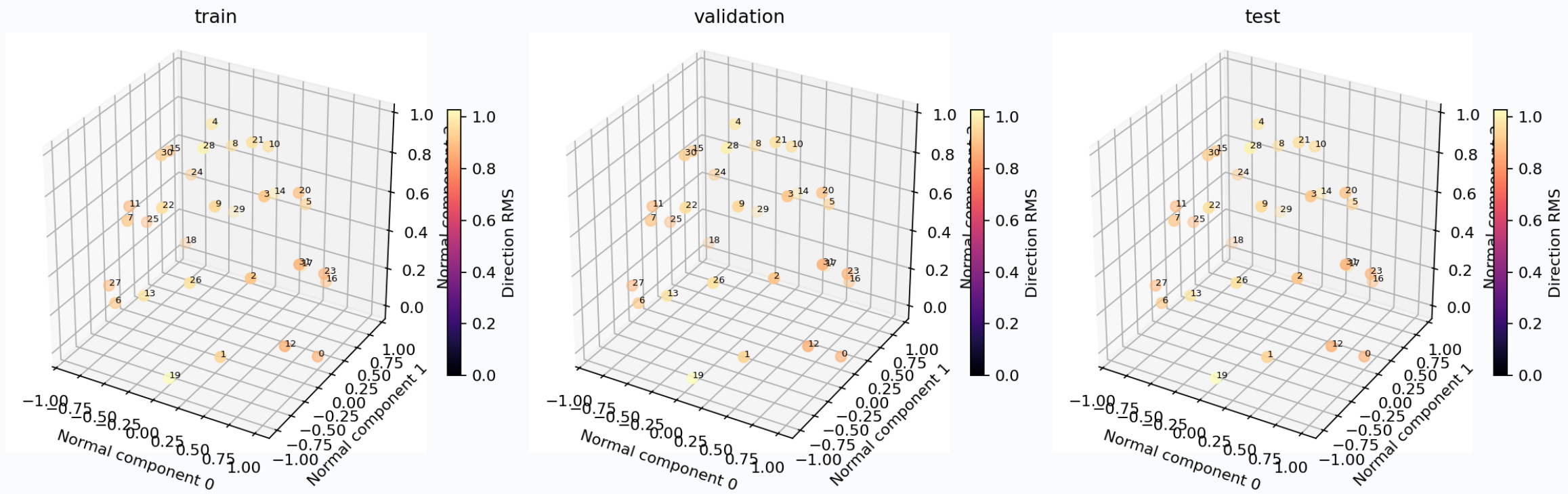


Figure 18 | 06 width robustness

sinogram_atlas / 06_width_robustness

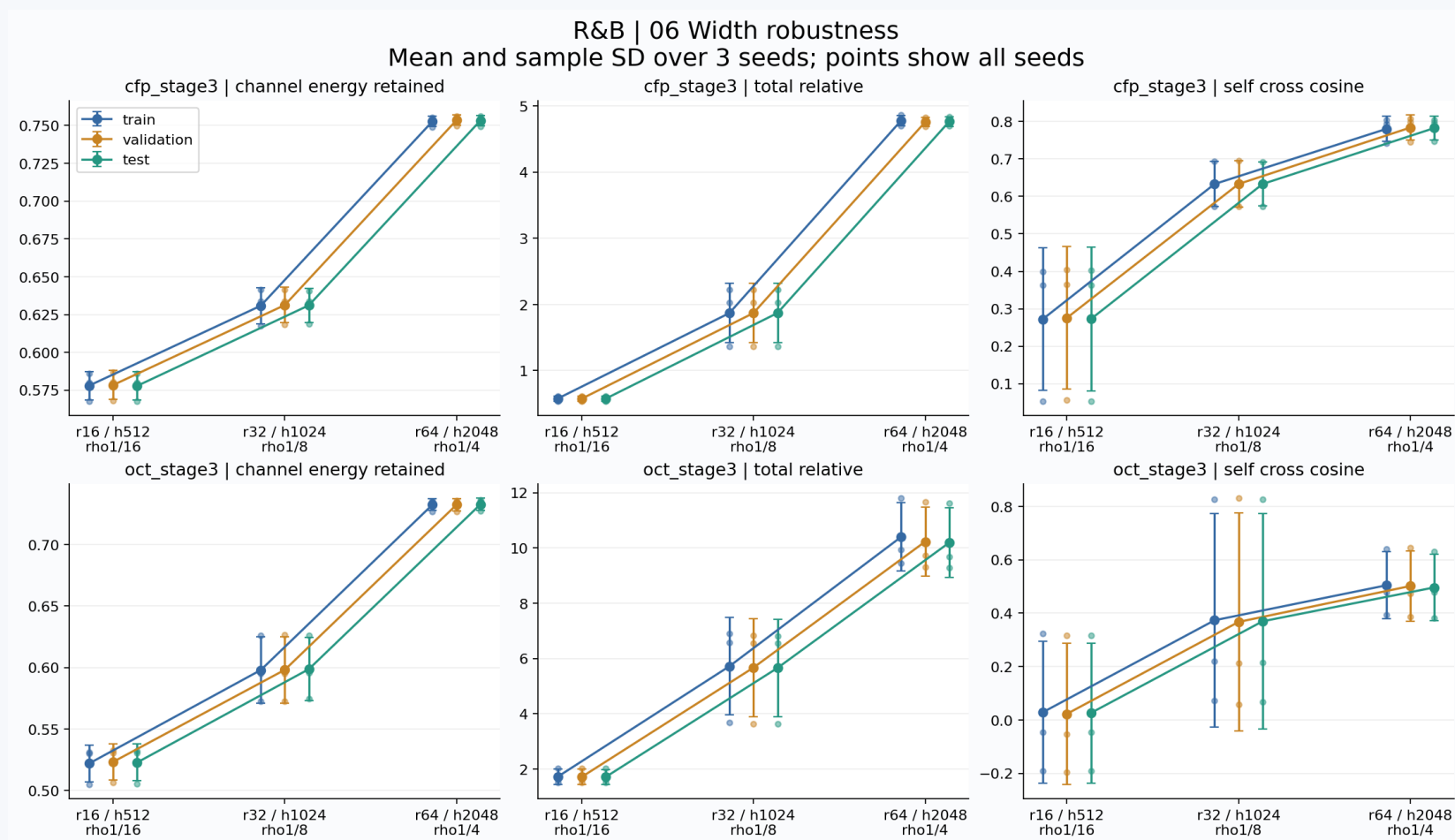


Figure 19 | 07 basis and ordinary communication

sinogram_atlas / 07_basis_and_ordinary_communication

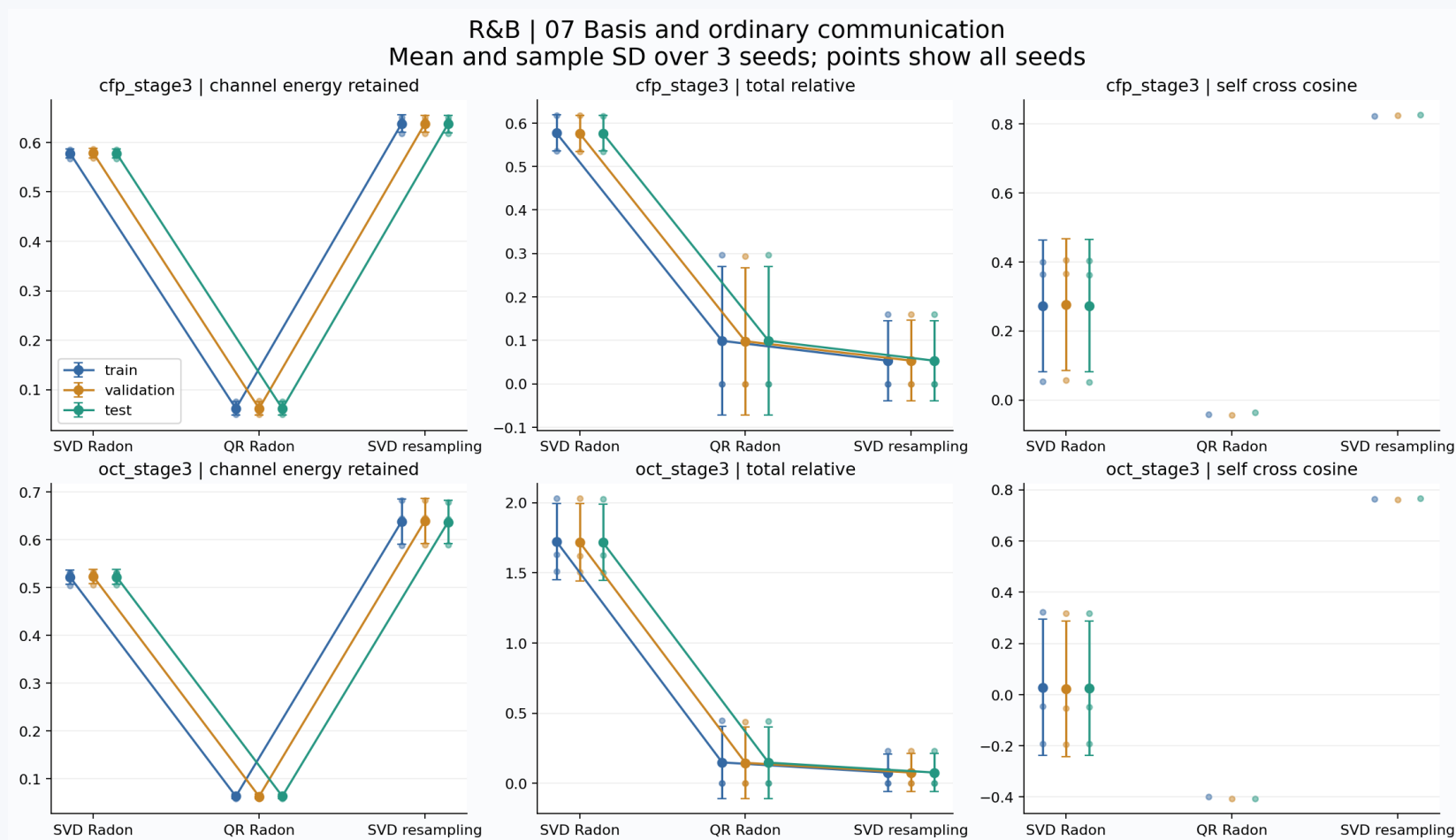


Figure 20 | 08 s axis spectra

sinogram_atlas / 08_s_axis_spectra

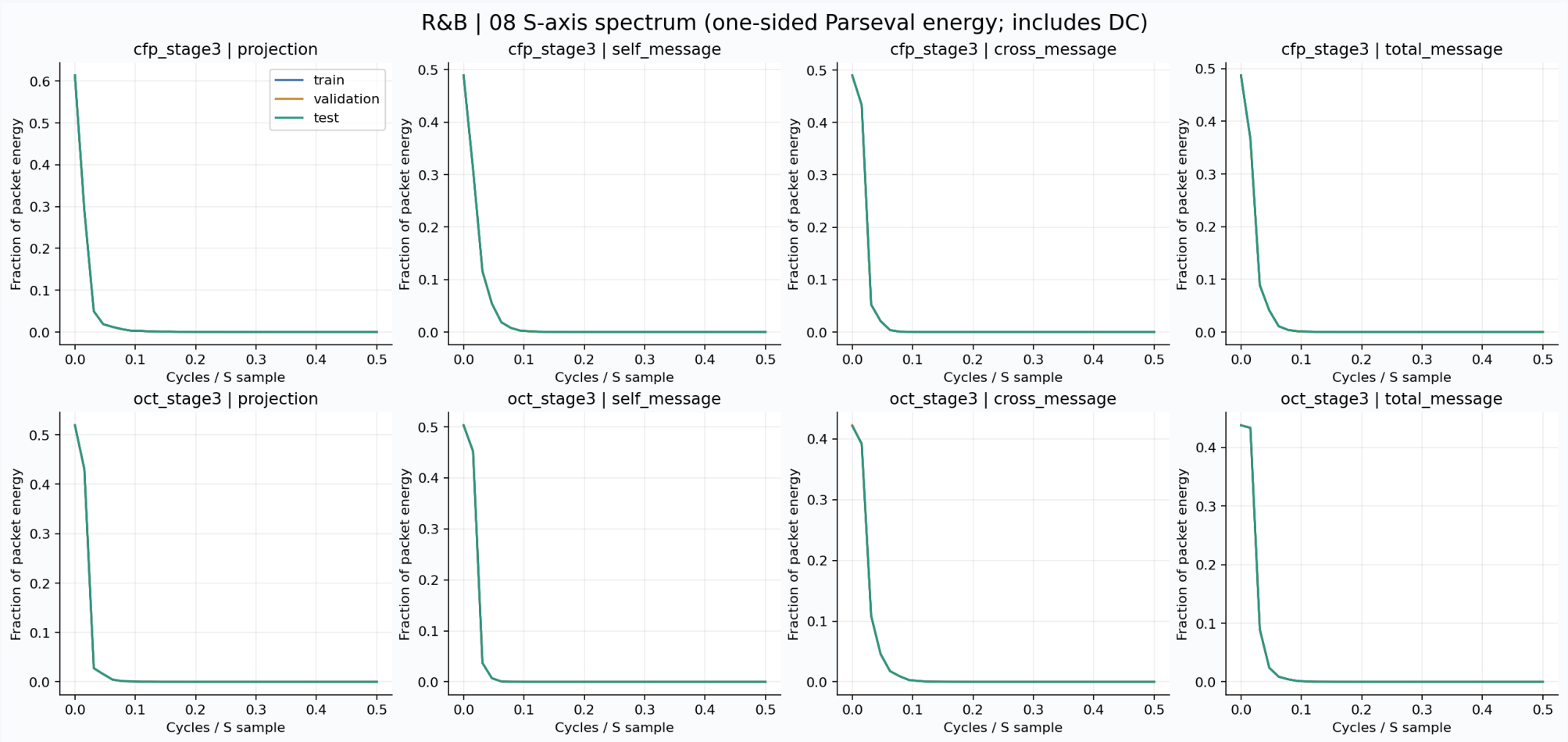


Figure 21 | 09 multidepth returns

sinogram_atlas / 09_multidepth_returns

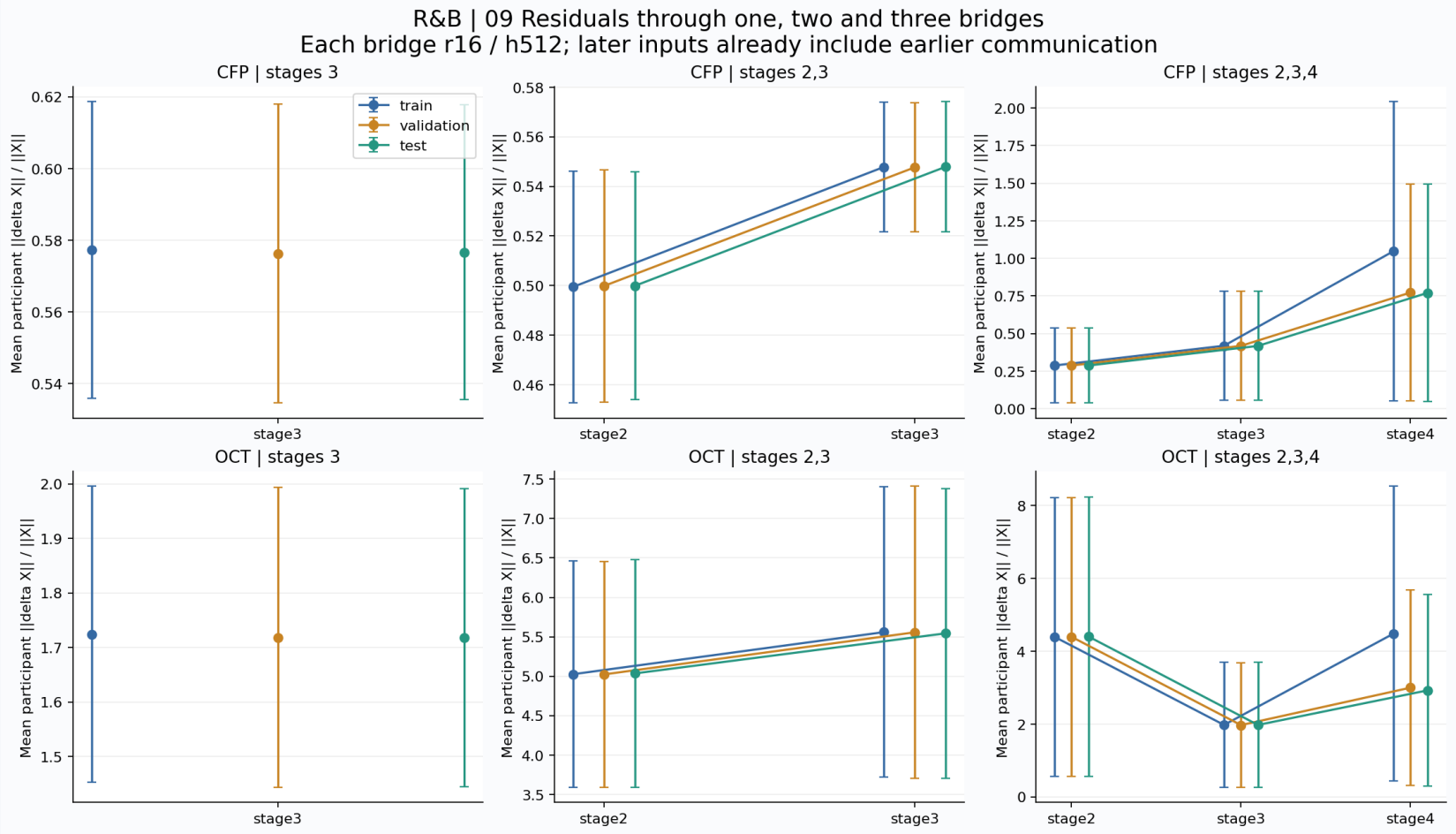


Figure 22 | 10 initial selected inputs

sinogram_atlas / 10_initial_selected_inputs

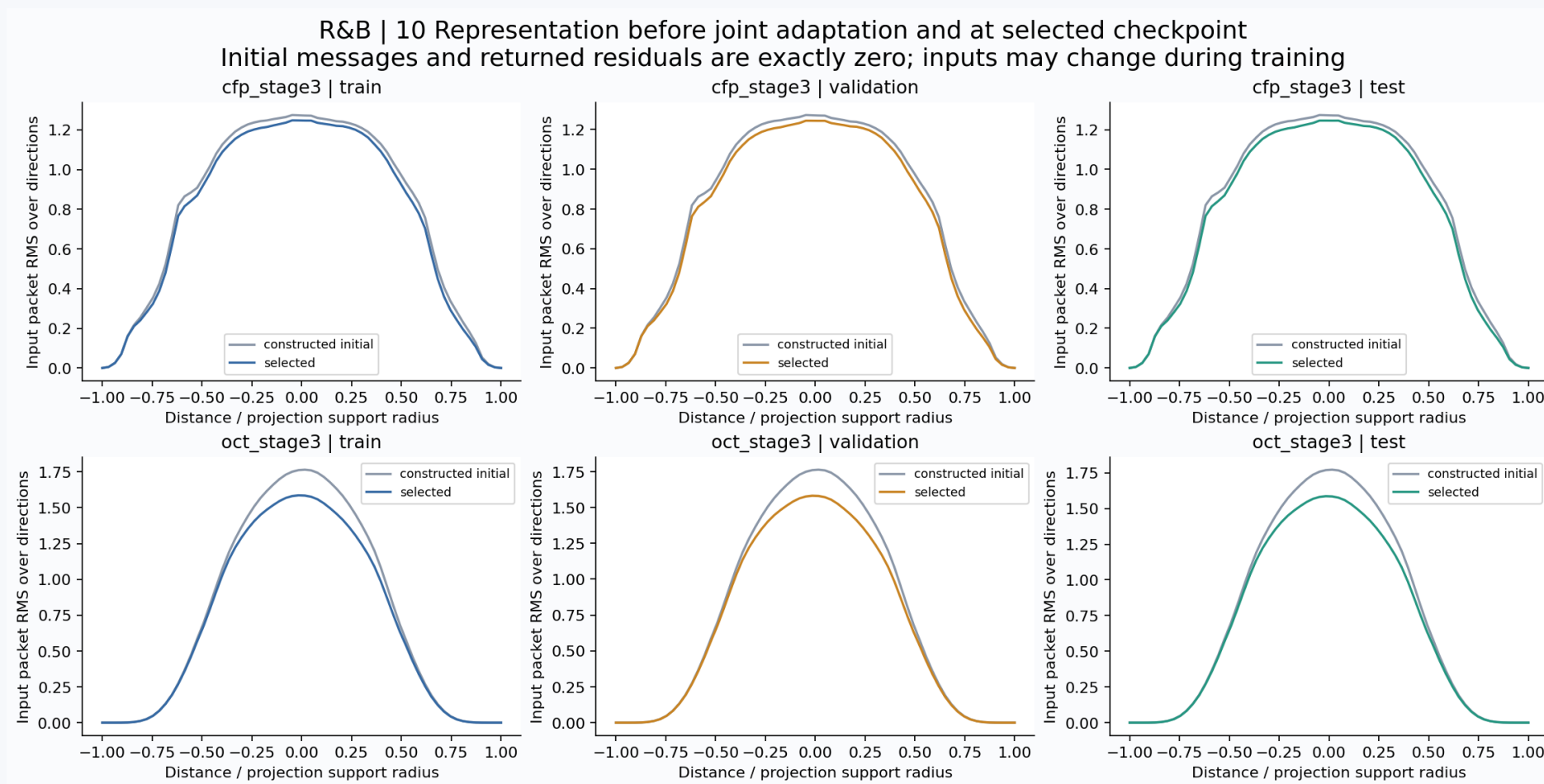


Figure 23 | BA cfp stage2

fourier_diagnostics / BA_cfp_stage2

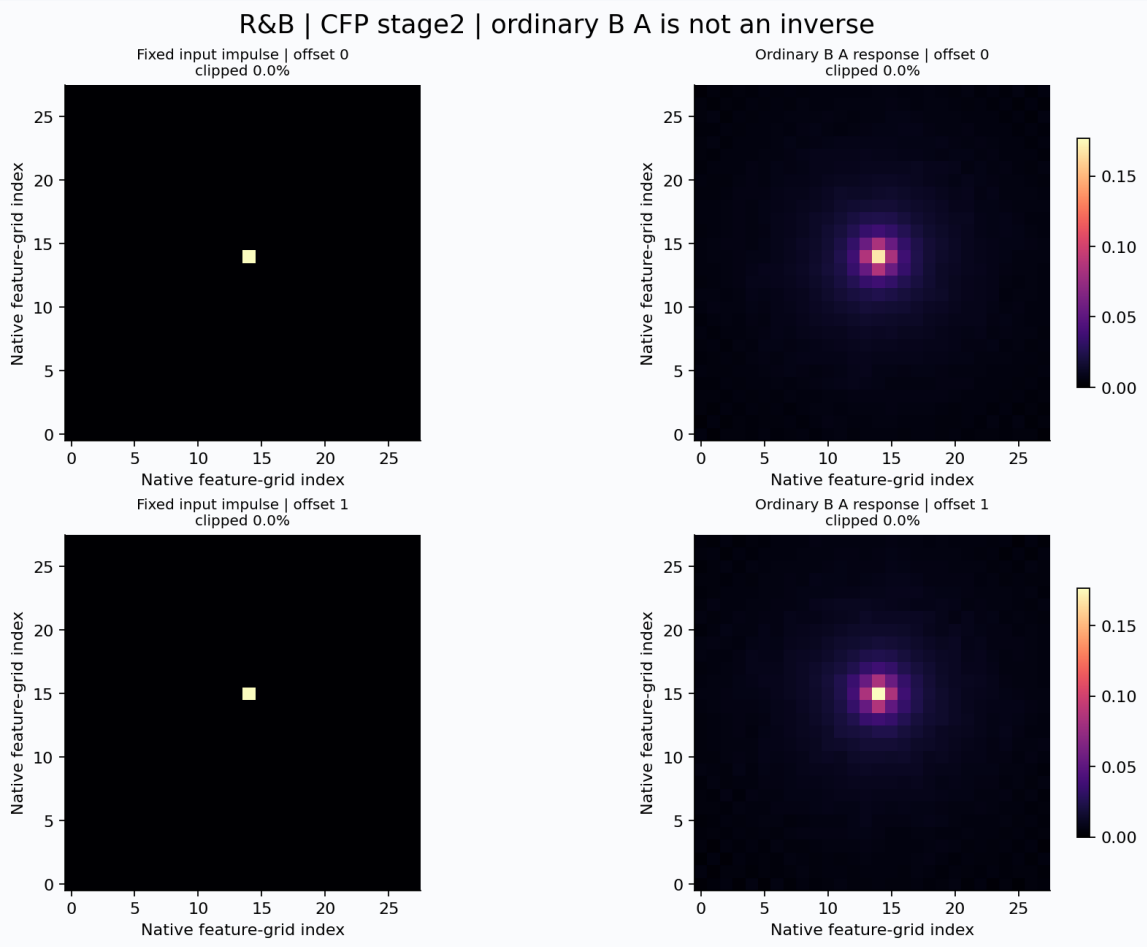


Figure 24 | BA cfp stage3

fourier_diagnostics / BA_cfp_stage3

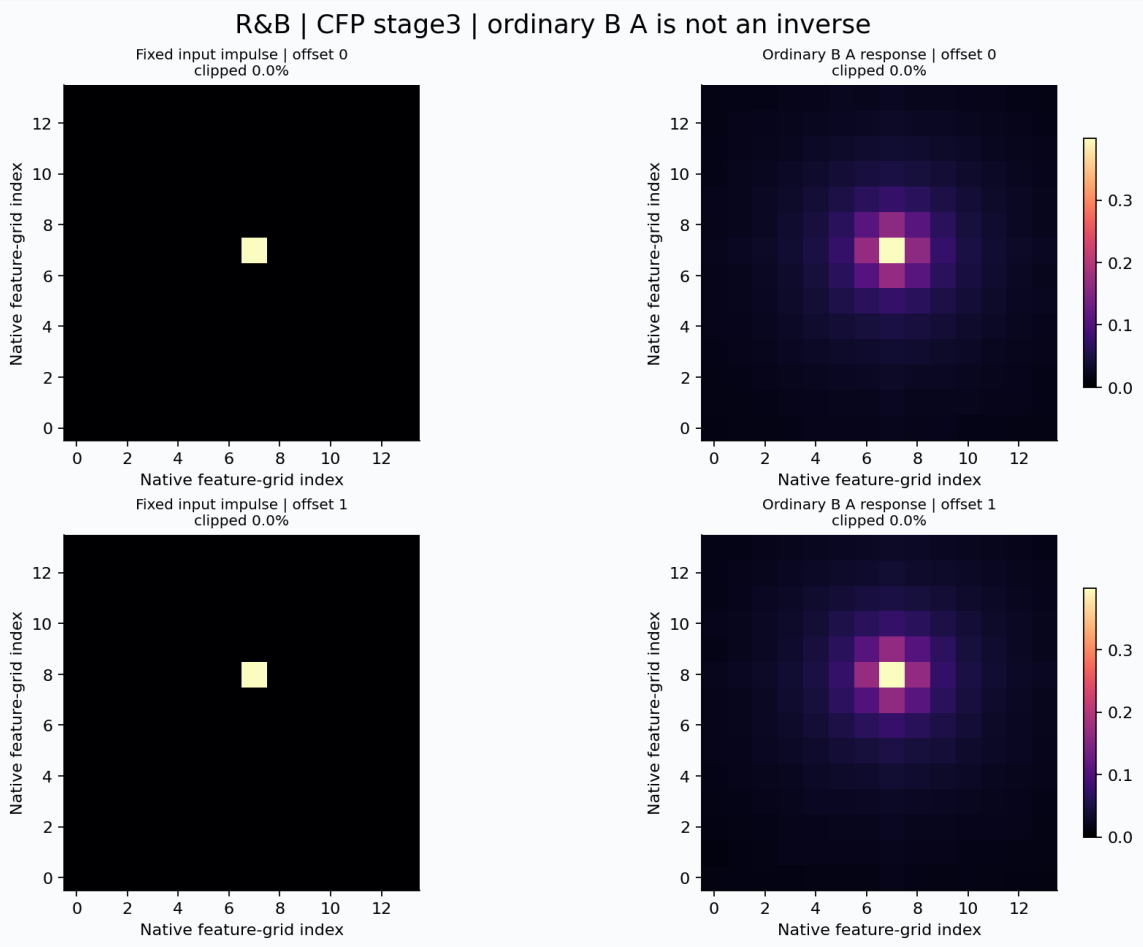


Figure 25 | BA cfp stage4

fourier_diagnostics / BA_cfp_stage4

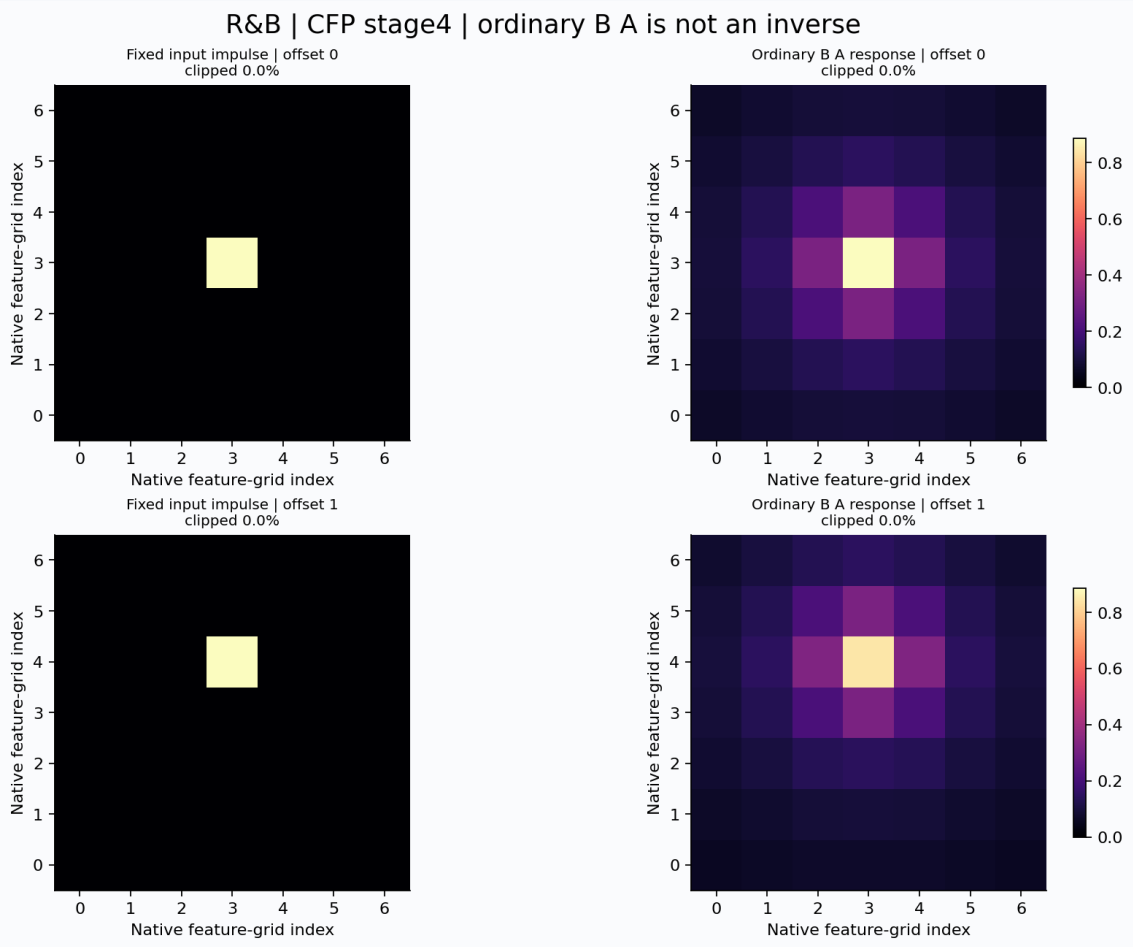


Figure 26 | BA oct stage2

fourier_diagnostics / BA_oct_stage2

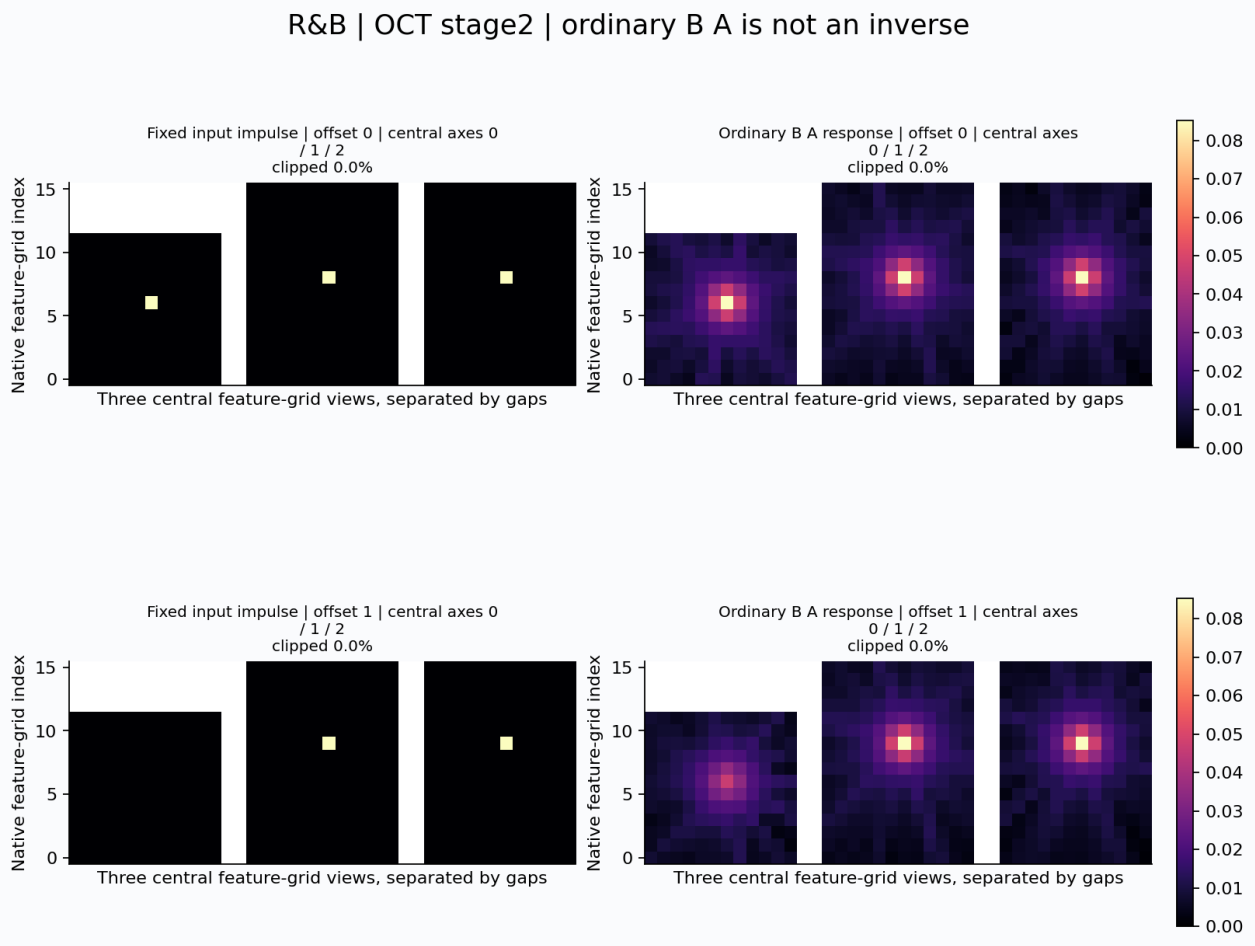


Figure 27 | BA oct stage3

fourier_diagnostics / BA_oct_stage3

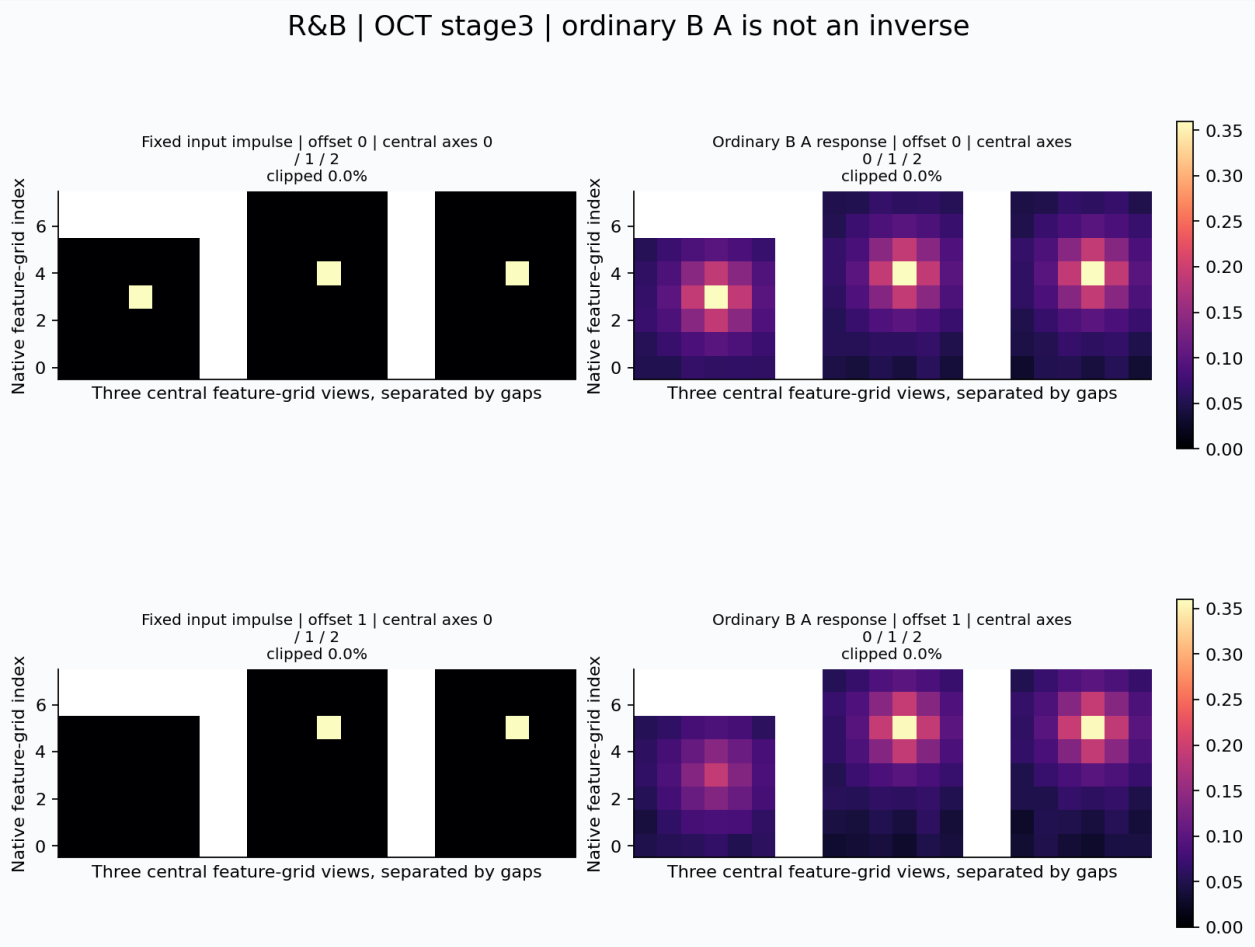


Figure 28 | BA oct stage4

fourier_diagnostics / BA_oct_stage4

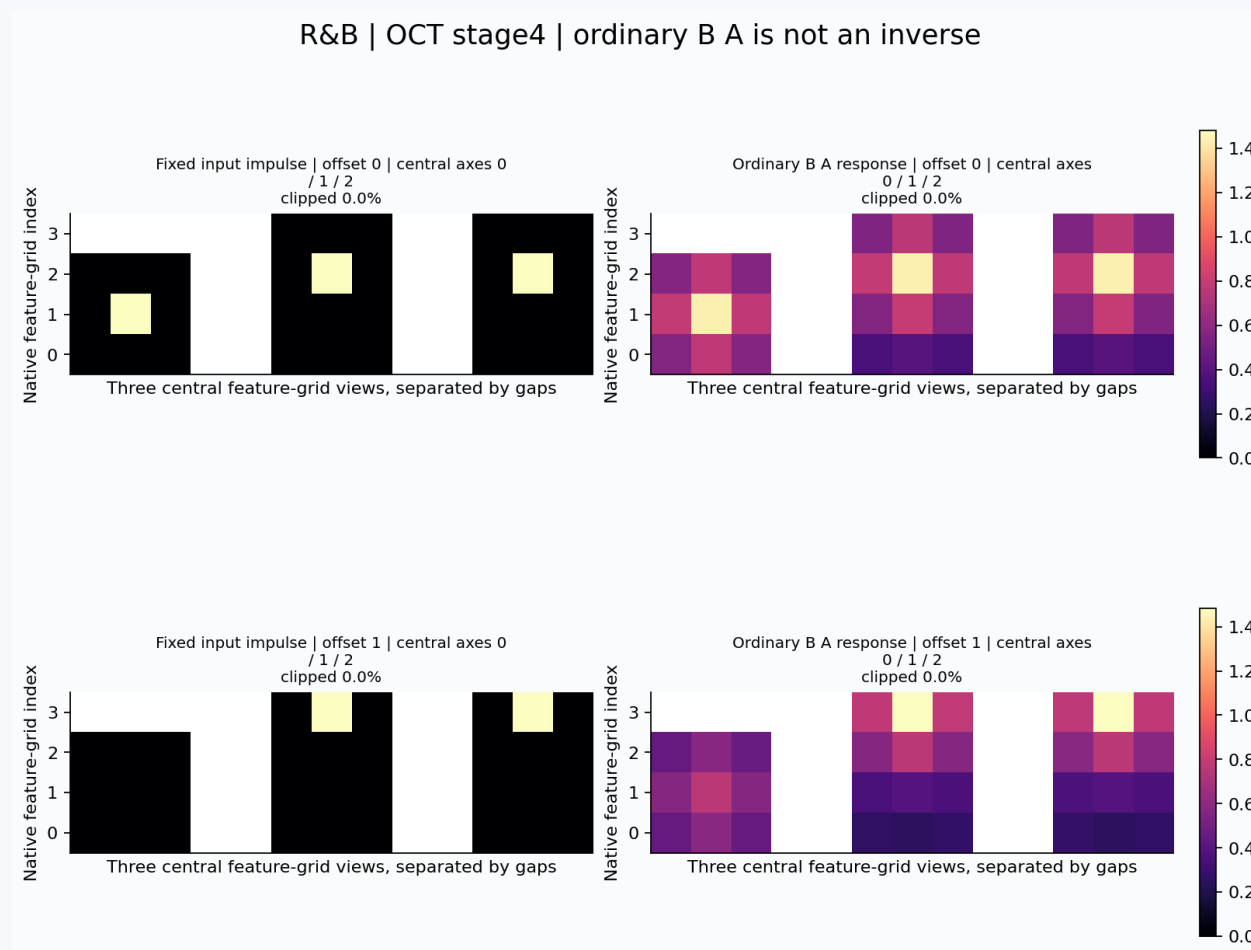


Figure 29 | fourier feature probes

fourier_diagnostics / fourier_feature_probes

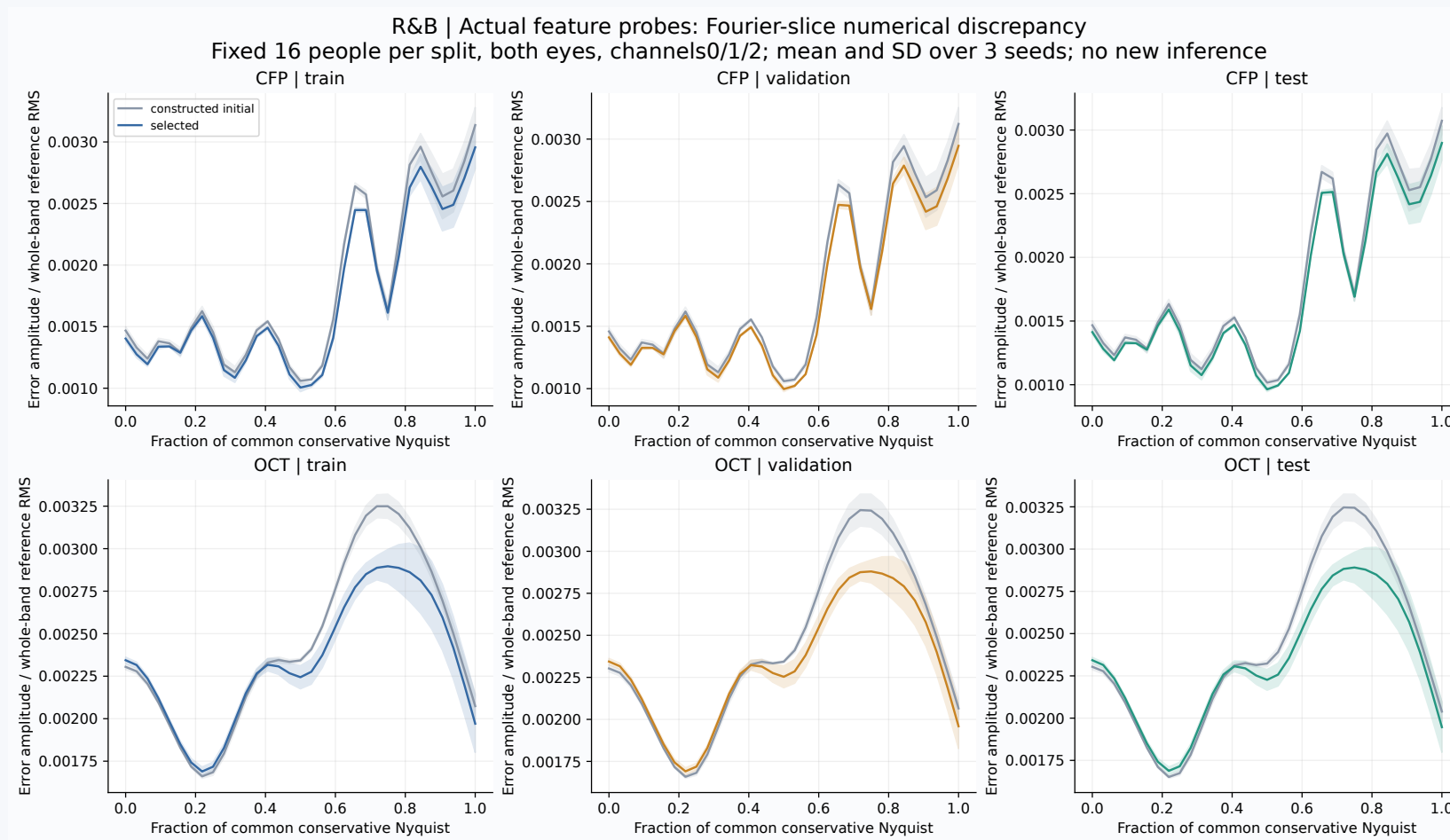


Figure 30 | fourier synthetic errors

fourier_diagnostics / fourier_synthetic_errors

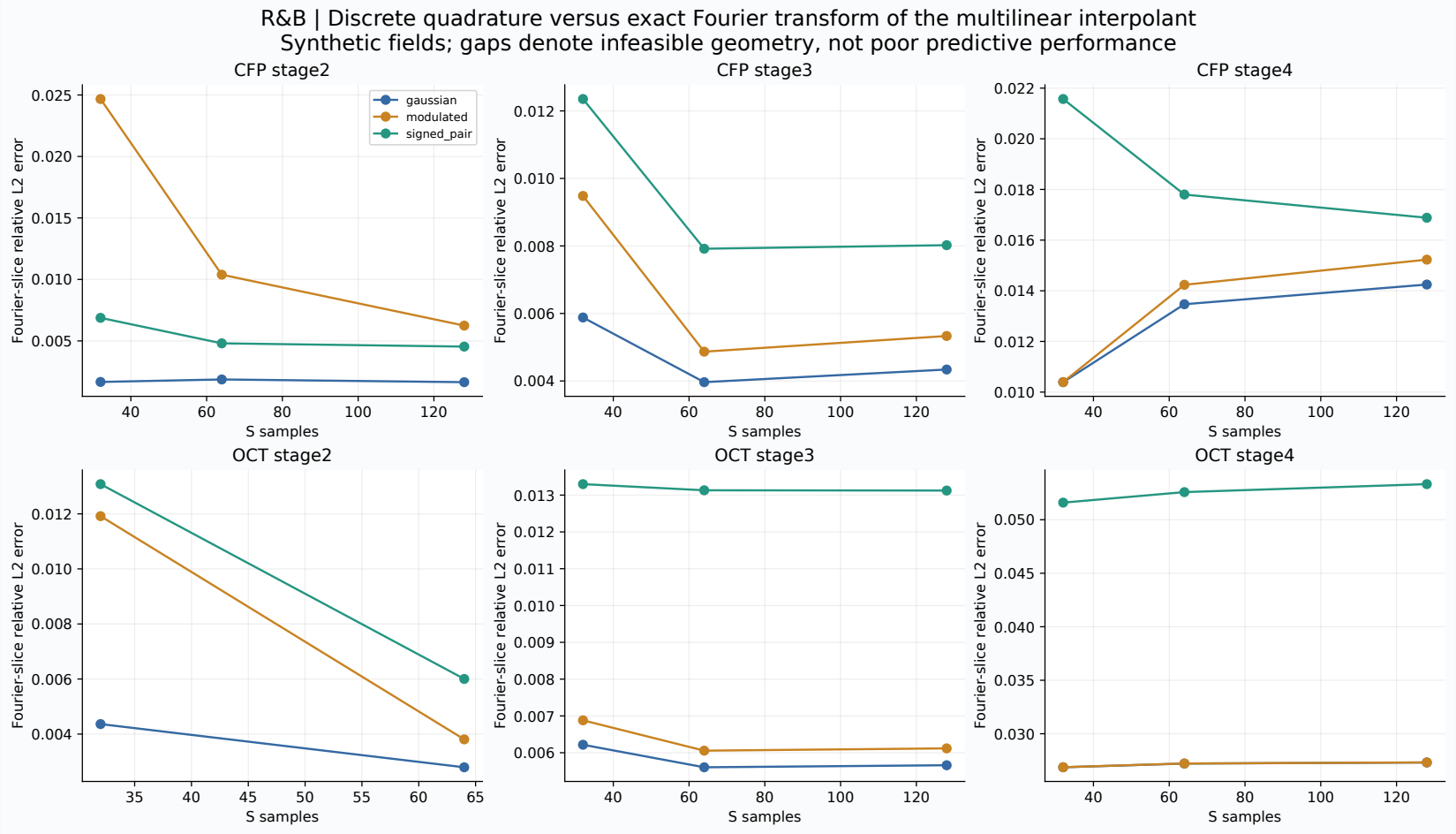


Figure 31 | kernel frequency gains

fourier_diagnostics / kernel_frequency_gains

